# Note

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## 1 ATLAS latest study

The couplings of longitudinally polarized $W$ and $Z$ bosons to the Higgs boson prevent the divergence of tree-level Vector Boson Scattering (VBS) amplitudes at high energies when the two outgoing bosons are longitudinally polarized, restoring unitarity at the TeV scale. Thus, the VBS processes with longitudinal vector bosons provide a sensitive test of the electroweak (EW) symmetry-breaking mechanism.

The most recent ATLAS study is detailed in Ref. [1]. This study focuses on VBS. Specifically, it investigates the EW production of same-sign $W$ boson pairs (EW $W^\pm W^\pm jj$), a process to which VBS provides a significant contribution. The key concepts and event selection criteria of this work are summarized below.

The object selection criteria are as follows:

- **Electrons:** Must have transverse momentum $p_{\mathrm{T}} > 27$ GeV and pseudorapidity $|\eta| < 2.47$, excluding the calorimeter transition region of $1.37 < |\eta| < 1.52$.
- **Muons:** Must have $p_{\mathrm{T}} > 27$ GeV and $|\eta| < 2.5$.
- **Jets:** At least two jets with $|\eta| < 4.5$ are required in each event. The leading jet must have $p_{\mathrm{T}} > 65$ GeV, and the subleading jet must have $p_{\mathrm{T}} > 35$ GeV.

The Signal Region (SR) event selection requires:

- A same-sign lepton pair with an invariant mass $m_{\ell\ell} > 20$ GeV.
- Missing transverse momentum magnitude, $E_{\mathrm{T}}^{\mathrm{miss}}$, greater than 30 GeV.
- The two highest-$p_{\mathrm{T}}$ jets must satisfy a dijet invariant mass $m_{jj} > 500$ GeV with an absolute rapidity difference $|\Delta y_{jj}| > 2.0$.

# 2 Monte Carlo simulation

## 2.1 Sample Generation

The signal processes are the electroweak (EW) production of $W^{\pm}W^{\pm}jj$ in various polarization states. We simulate the EW same-sign $W$ boson pair production at a center-of-mass energy of $\sqrt{s} = 13$ TeV. The events are generated using `MadGraph 3.3.1` [2]. Both $W$ bosons must decay leptonically. Parton showering and hadronization are simulated with `Pythia 8.306` [3], followed by a fast detector simulation using `Delphes 3.4.2` [4]. Jets are reconstructed via the anti-$k_t$ algorithm [5] with a distance parameter of $R = 0.4$, implemented in `FastJet 3.3.2` [6]. Reconstructed jets are required to have a transverse momentum of $p_{\mathrm{T}} > 25$ GeV.

The following `MadGraph` scripts are used to generate the Monte Carlo (MC) samples for these polarized processes:

**Longitudinal mode:** $pp \to W^{\pm}_{\mathbf{L}}W^{\pm}_{\mathbf{L}}jj$

```
generate p p > j j w+{0} w+{0} QCD=0, w+ > l+ vl
add process p p > j j w-{0} w-{0} QCD=0, w- > l- vl~
output EW_WWjj_LL
launch EW_WWjj_LL

shower=Pythia8
detector=Delphes
analysis=OFF
madspin=OFF
done

Cards/delphes_card.dat

set run_card nevents 10000
set run_card ebeam1 6500.0
set run_card ebeam2 6500.0

set run_card ptj 34.0
set run_card ptl 26.0
set run_card misset 29.0
```

```
set run_card etaj 4.6
set run_card etal 2.6

set run_card mmjj 480
set run_card deltaeta 1.9

set run_card ptj1min 64
set run_card ptj2min 34

set run_card use_syst False

done
```

**Transverse mode:** $pp \to W^{\pm}_{\mathbf{T}} W^{\pm}_{\mathbf{T}} jj$

```
generate p p > j j w+{T} w+{T} QCD=0, w+ > l+ vl
add process p p > j j w-{T} w-{T} QCD=0, w- > l- vl~
...
```

**Mixed mode:** $pp \to W^{\pm}_{\mathbf{L}} W^{\pm}_{\mathbf{T}} jj$

```
generate p p > j j w+{0} w+{T} QCD=0, w+ > l+ vl
add process p p > j j w-{0} w-{T} QCD=0, w- > l- vl~
...
```

## 2.2 Event Selection

The event selection criteria applied after the Delphes simulation are defined as follows:

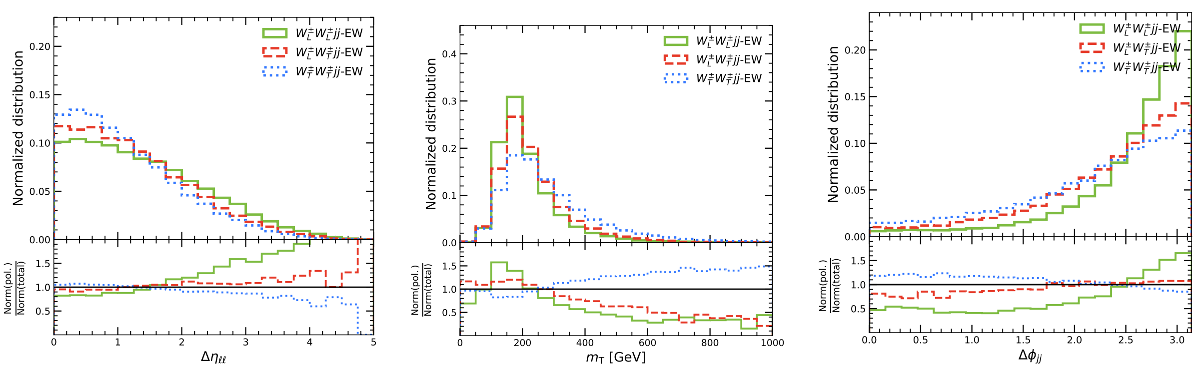
- **Lepton cut:** The invariant mass of the same-sign lepton pair, $m_{\ell\ell}$, must be greater than 20 GeV.
- **MET cut:** The missing transverse momentum magnitude, $E_{\mathrm{T}}^{\mathrm{miss}}$, must be greater than 30 GeV.
- **Jet cut:** The two highest-$p_{\mathrm{T}}$ tagging jets must satisfy a dijet invariant mass of $m_{jj} >$ 500 GeV with an absolute rapidity difference of $|\Delta y_{jj}| > 2.0$.

Table 1 summarizes the cutflow (number of expected events and the corresponding selection efficiencies) at each selection stage. The reconstruction requirements for electrons, muons, and jets are identical to those described in Section 1.

Table 1: Number of events and selection efficiencies (pass rates) for EW $W^{\pm}W^{\pm}jj$ production at different selection stages.

| Cut | LL | pass rate | LT | pass rate | TT | pass rate |
|---|---|---|---|---|---|---|
| Total | 100000 | 1.00 | 100000 | 1.00 | 100000 | 1.00 |
| Lepton cut | 26777 | 0.27 | 27270 | 0.27 | 29825 | 0.30 |
| MET cut | 24781 | 0.25 | 25518 | 0.26 | 28323 | 0.28 |
| Jet cut | 20005 | 0.20 | 21087 | 0.21 | 23298 | 0.23 |

Figure 1 displays the kinematic distributions for the various polarization states in the SR. Specifically, we present the absolute pseudorapidity difference between the leading and subleading leptons $|\Delta\eta_{\ell\ell}|$, the transverse mass of the dilepton and $E_{\mathrm{T}}^{\mathrm{miss}}$ system $m_{\mathrm{T}}$, and the azimuthal angle difference between the leading and subleading jets $\Delta\phi_{jj}$. The ratios of the individual process contributions to the total signal are shown in the bottom panels. For these combined distributions, we assume that the $W_{\mathrm{L}}^{\pm}W_{\mathrm{L}}^{\pm}jj$, $W_{\mathrm{L}}^{\pm}W_{\mathrm{T}}^{\pm}jj$, and $W_{\mathrm{T}}^{\pm}W_{\mathrm{T}}^{\pm}jj$ states contribute 10%, 30%, and 60% to the total signal, respectively.




(a) $|\Delta\eta_{\ell\ell}|$ distribution (b) $m_{\mathrm{T}}$ distribution (c) $\Delta\phi_{jj}$ distribution

Figure 1: Kinematic distributions of $|\Delta\eta_{\ell\ell}|$, $m_{\mathrm{T}}$, and $\Delta\phi_{jj}$ in the signal region. The ratios of the contributions from different polarization states are shown in the bottom panels.

### 2.3 Absolute Cross-Sections

Table 2 summarizes the absolute cross-sections for the EW $W^{\pm}W^{\pm}jj$ production across different selection stages. To validate our simulation framework, the final cross-sections after the jet cut are compared with the reference results from a previous thesis [7].

Table 2: Absolute cross-sections (in fb) for the EW $W^{\pm}W^{\pm}jj$ production at various selection stages. The reference values from the previous thesis and the relative fractions are provided at the final jet cut stage for comparison.

| Cut | LL | LT | TT | Total |
|---|---|---|---|---|
| Total | 0.443 | 2.514 | 3.824 | 6.781 |
| Lepton cut | 0.119 | 0.685 | 1.140 | 1.945 |
| MET cut | 0.110 | 0.641 | 1.083 | 1.834 |
| Jet cut | 0.089 | 0.530 | 0.891 | 1.510 |
| Jet cut (Thesis) | 0.177 | 1.030 | 1.718 | 2.925 |
| Fraction | 5.9% | 35.1% | 59.0% | 100% |
| Fraction (Thesis) | 6.1% | 35.2% | 58.7% | 100% |

As observed in Table 2, while the relative fractions of the polarization states are in excellent agreement with the reference thesis, the absolute cross-sections in our simulation are approximately a factor of two lower. Further investigation is needed.

### 2.4 $WW$ Center-of-Mass Reference Frame

To properly study the scattering dynamics, we modify our `MadGraph` scripts to define the $W$ boson polarizations in the $W^{\pm}W^{\pm}$ center-of-mass (CM) frame.

The following generation commands are used to produce the MC samples with this specific reference frame definition:

**Longitudinal mode:** $pp \to W^{\pm}_{\mathbf{L}}W^{\pm}_{\mathbf{L}}jj$

```
generate p p > w+{0} w+{0} j j QCD=0, w+ > l+ vl
add process p p > w-{0} w-{0} j j QCD=0, w- > l- vl~
output MG5/EW_WWjj_LL-WW_cmf
launch MG5/EW_WWjj_LL-WW_cmf

shower=Pythia8
detector=Delphes
analysis=OFF
madspin=OFF
done
```

```
Cards/delphes_card.dat

set run_card nevents 10000
set run_card ebeam1 6500.0
set run_card ebeam2 6500.0

set run_card ptj 34.0
set run_card etaj 4.6
set run_card mmjj 480
set run_card deltaeta 1.9

set run_card etal 2.6
set run_card ptl 26.0
set run_card misset 29.0

set run_card ptj1min 64
set run_card ptj2min 34

set run_card me_frame 3, 4, 5, 6

set run_card use_syst False

done
```

**Transverse mode:** $pp \to W^{\pm}_{\mathbf{T}} W^{\pm}_{\mathbf{T}} jj$

```
generate p p > w+{T} w+{T} j j QCD=0, w+ > l+ vl
add process p p > w-{T} w-{T} j j QCD=0, w- > l- vl~
...
```

**Mixed mode:** $pp \to W^{\pm}_{\mathbf{L}} W^{\pm}_{\mathbf{T}} jj$

```
generate p p > w+{0} w+{T} j j QCD=0, w+ > l+ vl
add process p p > w-{0} w-{T} j j QCD=0, w- > l- vl~
...
```

The key modification lies in the `me_frame` setting. More details can be found in ref. [8]. It is crucial to note that when utilizing the in-line decay syntax, `MadGraph` expands the decay products directly into the final-state particle list. Consequently, the indices must be specified as `3, 4, 5, 6` to encompass all the decay products (leptons and neutrinos) required to reconstruct the $WW$ system.

The remaining shower, detector, and run card settings for the transverse and mixed modes are identical to those of the longitudinal mode and are omitted here for brevity.

Table 3 summarizes the absolute cross-sections for the EW $W^\pm W^\pm jj$ production when the polarizations are defined in the $W^\pm W^\pm$ CM frame. The absolute cross-sections are still a factor of two lower.

Table 3: Absolute cross-sections for the EW $W^\pm W^\pm jj$ production at various selection stages, with $W$ boson polarizations defined in the $W^\pm W^\pm$ center-of-mass frame. Reference values from the previous thesis and the relative fractions are provided at the final jet-cut stage for comparison.

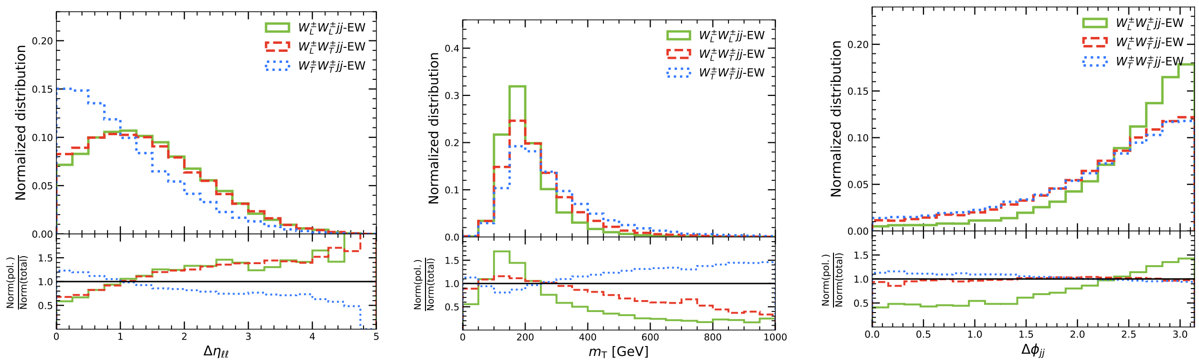
| Cut | LL [fb] | LT [fb] | TT [fb] | Total [fb] |
|---|---|---|---|---|
| Total | 0.677 | 2.190 | 3.889 | 6.756 |
| Lepton cut | 0.190 | 0.577 | 1.143 | 1.910 |
| MET cut | 0.177 | 0.535 | 1.081 | 1.793 |
| Jet cut | 0.147 | 0.443 | 0.895 | 1.485 |
| Jet cut (Thesis) | 0.277 | 0.874 | 1.744 | 2.869 |
| Fraction | 9.4% | 30.4% | 60.2% | 100% |
| Fraction (Thesis) | 9.6% | 30.6% | 59.8% | 100% |

Figure 2 displays the kinematic distributions for the various polarization states in the SR within the $WW$ center-of-mass frame. These results are consistent with Figure 2 in ref. [1].

## 2.5 Simulation with Sherpa

To investigate the cross-section discrepancy, we employ a different Monte Carlo generator, `Sherpa 3` [9], which is also the primary generator utilized in the ATLAS analysis.

We simulate the EW same-sign $W$ boson pair production at a center-of-mass energy of $\sqrt{s} = 13$ TeV. Both $W$ bosons must decay leptonically. Parton showering and hadronization are handled internally by `Sherpa`, followed by a fast detector simulation using `Delphes 3.4.2` [4]. Jets are reconstructed via the anti-$k_t$ algorithm [5] with a distance parameter of $R = 0.4$,




(a) $|\Delta\eta_{\ell\ell}|$ distribution (b) $m_{\mathrm{T}}$ distribution (c) $\Delta\phi_{jj}$ distribution

Figure 2: Kinematic distributions of $|\Delta\eta_{\ell\ell}|$, $m_{\mathrm{T}}$, and $\Delta\phi_{jj}$ in the signal region. The ratios of the contributions from different polarization states are shown in the bottom panels.

implemented in `FastJet 3.3.2` [6]. The reconstructed jets are required to have a transverse momentum of $p_{\mathrm{T}} > 25$ GeV.

Unlike `MadGraph`, which can generate pure polarized samples, `Sherpa` generates events and assigns polarization weights to each of them. These weights represent the probability of an event belonging to a specific polarization state. Due to technical constraints within the generator, the $W^+W^+jj$ and $W^-W^-jj$ processes must be generated as separate samples.

The following YAML configuration snippet illustrates the generation of the $W^+W^+jj$ sample using `Sherpa`:

**$W^+W^+jj$ sample**

```
EVENTS: 100
RANDOM_SEED: 1

BEAMS: 2212
BEAM_ENERGIES: 6500

PDF_SET: NNPDF30_nlo_as_0118

ME_GENERATORS:
  - Comix
COMIX_DEFAULT_GAUGE: 0

# scales
```

```
MEPS:
  CORE_SCALE: VAR{0.5*Abs2(p[2]+p[3])}

# physical constants and width settings
PARTICLE_DATA:
  24: {Width: 0} # W Boson
  23: {Width: 0} # Z Boson
WIDTH_SCHEME: Fixed

# decay and polarization settings
HARD_DECAYS:
  Enabled: true
  Channels:
    24,12,-11: {Status: 2} # W+ -> nu_e e+
    24,14,-13: {Status: 2} # W+ -> nu_mu mu+
  Pol_Cross_Section:
    Enabled: true
    Reference_System: [Lab, COM]

# process definition: same-sign W boson pair production with two jets
PROCESSES:
  - 93 93 -> 24 24 93 93:
      Order: {QCD: 0, EW: 4}

# final state particles cuts
SELECTORS:
  - [PT, 93, 34, 10000]
  - [Eta, 93, -4.6, 4.6]
  - [Mass, 93, 93, 480, 10000]
  - [DY, 93, 93, 1.9, 10000]

  - VariableSelector:
      Variable: PT
      Flavs: 93
      Ranges:
        - [60.0, 10000.0]
```

```
    Ordering: [PT_UP]

# output settings
EVENT_OUTPUT: HepMC3_GenEvent[EW_WpWpjj.hepmc]
```

**$W^-W^-jj$ sample**

```
EVENTS: 100
RANDOM_SEED: 1

BEAMS: 2212
BEAM_ENERGIES: 6500

PDF_SET: NNPDF30_nlo_as_0118

ME_GENERATORS:
  - Comix
COMIX_DEFAULT_GAUGE: 0

# scales
MEPS:
  CORE_SCALE: VAR{0.5*Abs2(p[2]+p[3])}

# physical constants and width settings
PARTICLE_DATA:
  24: {Width: 0} # W Boson
  23: {Width: 0} # Z Boson
WIDTH_SCHEME: Fixed

# decay and polarization settings
HARD_DECAYS:
  Enabled: true
  Channels:
    -24,-12,11: {Status: 2} # W- -> nubar e-
    -24,-14,13: {Status: 2} # W- -> nubar mu-
  Pol_Cross_Section:
```

```
    Enabled: true
    Reference_System: [Lab, COM]

# process definition: same-sign W boson pair production with two jets
PROCESSES:
  - 93 93 -> -24 -24 93 93:
      Order: {QCD: 0, EW: 4}

# final state particles cuts
SELECTORS:

  - [PT, 93, 34, 10000]
  - [Eta, 93, -4.6, 4.6]
  - [Mass, 93, 93, 480, 10000]
  - [DY, 93, 93, 1.9, 10000]

  - VariableSelector:
      Variable: PT
      Flavs: 93
      Ranges:
        - [60.0, 10000.0]
      Ordering: [PT_UP]

# output settings
EVENT_OUTPUT: HepMC3_GenEvent[EW_WmWmjj.hepmc]
```

It is important to note that the decay channel specification in `Sherpa` is strictly sensitive. The particle ordering must match the internal definitions precisely; for instance, `24,12,-11: {Status: 2}` is valid, whereas `24,-11,12: {Status: 2}` will not be recognized.

Table 4 details the absolute cross-sections for the individual $W^+W^+jj$ and $W^-W^-jj$ samples at various selection stages, with the polarization states defined in the $W^\pm W^\pm$ center-of-mass frame.

The combined absolute cross-sections for the EW $W^\pm W^\pm jj$ production are summarized in table 5. While the relative fractions of the polarization states remain in excellent agreement with the reference thesis, the combined absolute cross-section generated by `Sherpa`

Table 4: Absolute cross-sections for the separate $W^+W^+jj$ and $W^-W^-jj$ samples at various selection stages simulated with `Sherpa`. Polarization states are defined in the $W^\pm W^\pm$ center-of-mass frame.

| | $W^+W^+jj$ [fb] | | | | $W^-W^-jj$ [fb] | | | |
|---|---|---|---|---|---|---|---|---|
| Stage | LL | LT | TT | Total | LL | LT | TT | Total |
| Total | 0.527 | 1.633 | 2.949 | 5.108 | 0.162 | 0.475 | 0.850 | 1.487 |
| Lepton cut | 0.153 | 0.441 | 0.863 | 1.457 | 0.048 | 0.143 | 0.295 | 0.486 |
| MET cut | 0.132 | 0.394 | 0.776 | 1.302 | 0.040 | 0.122 | 0.250 | 0.412 |
| Jet cut | 0.106 | 0.319 | 0.633 | 1.057 | 0.030 | 0.094 | 0.199 | 0.323 |

(1.380 fb at the jet cut stage) is still approximately a factor of two lower than the reference value (2.933 fb).

Table 5: Combined absolute cross-sections for the EW $W^\pm W^\pm jj$ production simulated with `Sherpa`. Reference values from the previous thesis and the relative fractions are provided at the final jet-cut stage for comparison.

| Cut | LL [fb] | LT [fb] | TT [fb] | Total [fb] |
|---|---|---|---|---|
| Total | 0.688 | 2.108 | 3.799 | 6.595 |
| Lepton cut | 0.202 | 0.583 | 1.158 | 1.943 |
| MET cut | 0.172 | 0.516 | 1.026 | 1.714 |
| Jet cut | 0.136 | 0.413 | 0.831 | 1.380 |
| Jet cut (Thesis) | 0.285 | 0.909 | 1.778 | 2.933 |
| Fraction | 9.9% | 29.9% | 60.2% | 100% |
| Fraction (Thesis) | 9.6% | 30.6% | 59.8% | 100% |

## 2.6 Particle-Level Selection

Upon a detailed review of the reference thesis [7], we identified that their reported yields and cross-sections are evaluated at the particle level rather than the reconstructed level. To perform a fair comparison, we modified our event selection framework to operate on particle-level objects.

Specifically, the variables accessed in the `Delphes` analysis were updated as follows:

- **Leptons:** Reconstructed leptons (`Electron` and `Muon` collections) are replaced by generator-level stable final-state leptons from the `Particle` branch.

- **Missing Transverse Momentum:** The reconstructed missing transverse energy (`MissingET`) is replaced by the true missing transverse momentum (`GenMissingET`).

- **Jets:** Reconstructed jets (`Jet`) are replaced by truth-level jets (`GenJet`).

Furthermore, the kinematic selection criteria at the particle level are different from the reconstructed-level SR selection defined in Section 2.2. Based on the reference thesis, the particle-level selection is defined as follows:

- **Leptons:** Exactly two same-sign leptons with $p_{\mathrm{T}} > 27$ GeV. Muons must satisfy $|\eta| < 2.5$. Electrons must satisfy $|\eta| < 2.47$, excluding the transition region $1.37 \leq |\eta| \leq 1.52$, with an additional restriction of $|\eta| < 1.37$ for the $ee$ channel.

- **Dilepton Mass & $Z$-veto:** The dilepton invariant mass must be $m_{\ell\ell} > 20$ GeV. Additionally, a $Z$-boson veto is applied to the $ee$ channel, requiring $|m_{ee} - m_Z| > 15$ GeV to suppress background contributions.

- **Missing Transverse Momentum:** $E_{\mathrm{T}}^{\mathrm{miss}} \geq 30$ GeV.

- **Jets:** At least two jets are required. The leading and subleading jets must satisfy $p_{\mathrm{T}} > 65$ GeV and $p_{\mathrm{T}} > 35$ GeV, respectively. The dijet invariant mass threshold is relaxed to $m_{jj} \geq 200$ GeV, and the rapidity difference must be $\Delta y_{jj} > 2$.

The PDF sets are also updated to match the reference thesis [7]. We use the `NNPDF23_nlo_as_0119` PDF set [10] for `MadGraph` and `PDF4LHC21_40_pdfas` PDF set [11]for `Sherpa`.

The updated absolute cross-sections for the EW $W^{\pm}W^{\pm}jj$ production at the particle level are summarized in Table 6. Although the relative fractions of the polarization states remain in excellent agreement with the reference thesis, a systematic normalization discrepancy still exists in the absolute cross-sections.

The thesis also mentioned the overlapping removal strategies for the jet and lepton objects. However, it is not clear if they applied the overlapping removal and what kind of strategy they considered.

## 2.7 Background Processes

In this section, we focus on the primary background sources: the irreducible QCD $W^{\pm}W^{\pm}jj$ production and the reducible $W^{\pm}Zjj$ process.

Table 6: Combined absolute cross-sections for the EW $W^{\pm}W^{\pm}jj$ production simulated with `MadGraph` and `Sherpa` using **particle-level selection**. Reference values from the previous thesis and the relative fractions are provided at the final jet-cut stage for comparison.

| Cut | MadGraph [fb] | | | | Sherpa [fb] | | | |
|---|---|---|---|---|---|---|---|---|
| | LL | LT | TT | Total | LL | LT | TT | Total |
| Total | 0.383 | 1.188 | 2.185 | 3.755 | 0.729 | 2.252 | 3.989 | 6.970 |
| Lepton cut | 0.267 | 0.850 | 1.665 | 2.782 | 0.291 | 0.906 | 1.746 | 2.943 |
| MET cut | 0.260 | 0.834 | 1.636 | 2.731 | 0.240 | 0.786 | 1.522 | 2.548 |
| Jet cut | 0.205 | 0.623 | 1.129 | 1.957 | 0.188 | 0.569 | 1.021 | 1.777 |
| Jet cut (Thesis) | 0.277 | 0.874 | 1.744 | 2.895 | 0.285 | 0.909 | 1.778 | 2.972 |
| Fraction | 10.5% | 31.9% | 57.7% | 100% | 10.6% | 32.0% | 57.5% | 100% |
| Fraction (Thesis) | 9.6% | 30.2% | 60.2% | 100% | 9.6% | 30.6% | 59.8% | 100% |

### 2.7.1 QCD $W^{\pm}W^{\pm}jj$ Production

The QCD-induced same-sign $W$ boson pair production in association with two jets is an irreducible background, as its final state is identical to that of the signal. At tree level, this process is of order $\mathcal{O}(\alpha_S^2\alpha^4)$.

The following `MadGraph` scripts are used to generate the MC samples:

```
generate p p > w+ w+ j j QCD<=2 QED<=2, w+ > l+ vl
add process p p > w- w- j j QCD<=2 QED<=2, w- > l- vl~

...
```

### 2.7.2 INT $W^{\pm}W^{\pm}jj$ Production

The interference (INT) $W^{\pm}W^{\pm}jj$ background arises from the quantum interference between the purely electroweak and the QCD-induced amplitudes. At tree level, this process is of order $\mathcal{O}(\alpha_S\alpha^5)$.

### 2.7.3 $W^{\pm}Zjj$ Production

The $WZ$ production in association with jets, where $WZ \to \ell\nu\ell\ell$, is the most prominent reducible background. It mimics the same-sign $WW$ topology when one of the leptons from the $Z$ boson decay is "lost" (e.g., falls outside the detector acceptance or fails identification criteria).

The following `MadGraph` scripts are used to generate the MC samples:

```
generate p p > w+ z j j QCD=0, w+ > l+ vl, z > l+ l-
add process p p > w- z j j QCD=0, w- > l- vl~, z > l+ l-

...
```

### 2.7.4 Non-Prompt Leptons

Leptons originating from hadron decays and jets misidentified as leptons are referred to as non-prompt leptons. Since this background is estimated using data-driven methods in the experimental analysis, we omit its estimation in our current Monte Carlo simulation framework.

### 2.7.5 Full Process vs. Narrow Width Approximation (NWA)

The most rigorous approach to generating these events is to simulate the full $2 \to 6$ process. For instance, for the EW $W^{\pm}W^{\pm}jj$ signal, the full process command would be:

```
generate p p > j j l+ vl l+ vl QCD=0
add process p p > j j l- vl~ l- vl~ QCD=0
```

However, simulating the full process requires evaluating a vast number of Feynman diagrams and integrating over a highly complex phase space. This often leads to severe integration inefficiencies and divergences, making it prohibitive to generate a sufficient number of events.

We investigated the $W^{\pm}Zjj$ case and found that the full simulation and the Narrow Width Approximation (NWA) approach produce highly consistent absolute cross-sections after applying the SR cuts. Therefore, we temporarily adopt the NWA for our current generation setup to ensure efficiency.

### 2.7.6 Yields and Comparisons

To validate our simulation framework, the expected event yields are compared against the reference ATLAS results. Table 7 summarizes the expected signal and background yields in the SR. The "ATLAS" column represents the values reported in Table 3 of Ref. [12], the "Thesis" columns represent the values reported in Table 9.1 and Table G.2 of Ref. [7], while the remaining columns represent our simulation results. Here, we normalize our simulated yields to an integrated luminosity of $\mathcal{L} = 139\ \text{fb}^{-1}$.

Table 7: Expected yields in the signal region (SR). The ATLAS results are shown alongside Thesis values and our simulation estimates using different generator versions, assuming an integrated luminosity of 139 $\text{fb}^{-1}$.

| Process | ATLAS | Thesis 9.1 | Thesis G.2[1] | MG 3.3.1 | MG 3.7.0 | Sh 3.0.3 |
|---|---|---|---|---|---|---|
| $W^{\pm}W^{\pm}jj$ EW | 235 | 206.52 | 211.29[2] | 177.98 | 190.28 | 191.18 |
| $W^{\pm}W^{\pm}jj$ QCD | 24 | 24.05 | 24.27 | 14.38 | 12.96 | 19.12 |
| $W^{\pm}Zjj$ EW | 14.9 | 14.95 | 15.07 | 9.12 | 9.50 | 32.55 |
| $W^{\pm}Zjj$ QCD | 83 | 82.75[3] | 77.0 | 18.47 | 16.30 | 80.15 |

## 2.8 Determination of the Cross-Section for Sherpa Samples

When utilizing `Sherpa` to generate MC samples, there are two primary stages. First, the generator performs a numerical integration over the defined phase space to calculate the total cross-section. Subsequently, it generates events and writes the kinematic information and local cross-section into the resulting `HepMC` file.

In our previous analyses, the cross-section was estimated by extracting the values recorded event-by-event in the `HepMC` file and computing their average. However, the values written to the event record are local estimates derived during the generating stage. Consequently, they are subject to large statistical fluctuations, which can introduce biases into the final normalization.

Conversely, the cross-section reported in the `Sherpa` log output that was evaluated during the initial phase-space integration stage is more suitable. Because this integration stage utilizes vast sampling points before generating events, it reduces statistical fluctuations and yields a more stable cross-section.

Table 8 presents a comparison of the absolute cross-sections ($\sigma_{\text{gen}}$) and the corresponding expected yields ($N_{\text{exp}}$) in the SR obtained via these two methods: the integration log output (New) and the averaged `HepMC` weights (Old).

As observed in the table, the `HepMC` average consistently overestimates the expected yields by roughly 4% to 9% across various processes. All subsequent analyses will utilize the cross-section obtained from the `Sherpa` generation log.

[1]Values from the pre-fit event yields (Table G.2), where the three DNN score bins are merged.

[2]The sum of the LL, TL, and TT polarization processes.

[3]The predicted events appear to be scaled down (e.g., $82.75/2 \approx 41.38 \to 28.50$). The authors note in the caption that an additional normalization factor was applied to achieve a better agreement with the expected data. However, the exact derivation of this factor is not detailed.

Table 8: Comparison of the generation cross-sections ($\sigma_{\text{gen}}$) and expected yields ($N_{\text{exp}}$) in the SR using the Sherpa log output (New) versus the averaged HepMC values (Old). The percentage difference highlights the consistent overestimation bias introduced by the HepMC method.

| Sample | $\sigma_{\text{gen}}^{\text{New}}$ [fb] | $\sigma_{\text{gen}}^{\text{Old}}$ [fb] | $N_{\text{exp}}^{\text{New}}$ | $N_{\text{exp}}^{\text{Old}}$ | Diff. |
|---|---|---|---|---|---|
| $W^{\pm}W^{\pm}jj$ EW | 7.895 | 8.689 | 178.22 | 196.14 | −9.13% |
| $W^{\pm}W^{\pm}jj$ QCD | 2.245 | 2.441 | 17.88 | 19.44 | −8.01% |
| $W^{\pm}Zjj$ EW | 8.242 | 8.585 | 32.76 | 34.13 | −4.00% |
| $W^{\pm}Zjj$ QCD | 20.279 | 21.278 | 72.44 | 76.01 | −4.70% |

## 2.9 MadGraph Run Card Selection Criteria Issues

To increase the event generation efficiency for the SR phase space, we applied kinematic cuts at the generator level (as detailed in Section 2.4). While these cuts are appropriate for generating pure $W^{\pm}W^{\pm}jj$ signal samples, they introduce biases when extending the analysis to Control Regions (CR) or simulating reducible backgrounds such as $W^{\pm}Zjj$.

For the CR, the analysis investigates the low-$m_{jj}$ region, specifically 200 GeV $< m_{jj} <$ 500 GeV. The previous generator-level cut of `set run_card mmjj 480` is problematic, as it truncates almost all valid events that would populate this CR.

Furthermore, for the $W^{\pm}Zjj \rightarrow \ell\nu\ell\ell$ background, the global lepton transverse momentum cut `set run_card ptl 26.0` causes an underestimation. The `ptl` parameter in `MadGraph` enforces a minimum $p_{\text{T}}$ requirement on *all* generated leptons.

**Example:** Consider a $W^{+}Z \rightarrow \ell^{+}\nu\ell^{+}\ell^{-}$ event where the two same-sign leptons have $p_{\text{T}} = 45$ GeV and 35 GeV, while the third lepton ($\ell^{-}$) is soft with $p_{\text{T}} = 15$ GeV. In the detector simulation, this soft $\ell^{-}$ might be "lost" (e.g., falling outside the acceptance or failing identification), leaving exactly two same-sign leptons that pass the SR cuts ($p_{\text{T}} > 27$ GeV). However, because the third lepton's $p_{\text{T}}$ is lower than the 26.0 GeV threshold, `MadGraph` will discard this event at the generation level. Consequently, this topology is completely lost.

To resolve these issues, the `MadGraph` run card scripts must be updated to use looser, ordered cuts. Taking the QCD $W^{\pm}Zjj$ process as an example:

```
generate p p > w+ z j j QCD<=2 QED<=2, w+ > l+ vl, z > l+ l-
add process p p > w- z j j QCD<=2 QED<=2, w- > l- vl~, z > l+ l-
output MG5/WZjj_QCD-NWA-v370
launch MG5/WZjj_QCD-NWA-v370
```

```
shower=Pythia8
detector=Delphes
analysis=OFF
madspin=OFF
done

Cards/delphes_card.dat

set run_card nevents 10000
set run_card ebeam1 6500.0
set run_card ebeam2 6500.0

set run_card pdlabel lhapdf
set run_card lhaid 260000

set run_card ptj 20.0
set run_card etaj 5.0
set run_card mmjj 150.0
set run_card deltaeta 1.5
set run_card ptj1min 50.0
set run_card ptj2min 25.0

set run_card etal 2.7
set run_card ptl1min 20.0
set run_card ptl2min 20.0

set run_card misset 10.0

set run_card use_syst False
set run_card cut_decays True

done
```

The primary changes include relaxing the dijet invariant mass baseline to `mmjj 150.0`, and replacing the global `ptl` cut with ordered lepton cuts: `ptl1min 20.0` and `ptl2min 20.0`. This ensures that only the leading and subleading leptons are constrained, leaving the third lepton unrestricted. Other kinematic variables are relaxed to safely encompass the full reconstructed-level phase space.

Table 9 presents the updated event yields in the SR. In this revised comparison, we utilized the cross-sections from the `Sherpa` log outputs (Section 2.8) and applied the relaxed, ordered generator-level cuts for the `MadGraph` 3.7.0 samples to prevent phase-space truncation (Section 2.9).

Table 9: Updated expected yields in the SR. The ATLAS results are shown alongside the reference Thesis values and our refined simulation estimates (MG 3.7.0 with relaxed cuts and Sherpa 3.0.3 with log-based cross-sections). All values assume an integrated luminosity of 139 $\mathrm{fb}^{-1}$.

| Process | ATLAS | Thesis 9.1 | Thesis G.2 | MG 3.7.0 | Sherpa 3.0.3 |
|---|---|---|---|---|---|
| $W^{\pm}W^{\pm}jj$ EW | 235 | 206.52 | 211.29 | 204.08 | 178.22 |
| $W^{\pm}W^{\pm}jj$ QCD | 24 | 24.05 | 24.27 | 14.48 | 17.88 |
| $W^{\pm}Zjj$ EW | 14.9 | 14.95 | 15.07 | 22.94 | 32.76 |
| $W^{\pm}Zjj$ QCD | 83 | 82.75 | 77.0 | 50.23 | 72.44 |

# 3 Dense Neural Network

Because the kinematic relationships between the signal and background processes are complex, a single variable is insufficient for optimal separation. Therefore, multivariate analysis techniques, specifically Deep Neural Networks (DNNs), are employed to extract sensitive features to distinguish signal and background events [1]. We consider the Dense Neural Networks (Dense-NNs) as our first trial.

## 3.1 Input Features

The Dense-NN relies on a set of low-level and high-level kinematic variables. These input features are defined as follows:

- $(p_{\mathrm{T}}^{l}, \eta^{l}, \mathrm{flavor}^{l})$: Transverse momentum, pseudorapidity, and flavor type of the leading and sub-leading leptons.
- $(p_{\mathrm{T}}^{j}, \eta^{j})$: Transverse momentum and pseudorapidity of the leading and sub-leading jets.
- $E_{\mathrm{T}}^{\mathrm{miss}}$: Missing transverse momentum magnitude.

- $\Delta\phi(x, \ell_1)$: Difference in azimuthal angle between the object $x$ and the leading lepton. The object $x$ can be the sub-leading lepton, leading jet, sub-leading jet, or the missing transverse momentum $E_{\mathrm{T}}^{\mathrm{miss}}$.

- $Z_\ell^\star$: Zeppenfeld variables [13] of the leading and sub-leading leptons. The pseudorapidities of the leptons and tagging jets are used to define this variable:

$$Z_\ell^\star = \left| \frac{\eta_\ell - \frac{1}{2}\left(\eta_{j_1} + \eta_{j_2}\right)}{\eta_{j_1} - \eta_{j_2}} \right|. \tag{1}$$

- $m_{\mathrm{T}}(x, E_{\mathrm{T}}^{\mathrm{miss}})$: Transverse mass of the object $x$ and the $E_{\mathrm{T}}^{\mathrm{miss}}$ system. The object $x$ can be the leading lepton, sub-leading lepton, or the di-lepton system. The standard definition for a two-object system is:

$$m_{\mathrm{T}}(x, E_{\mathrm{T}}^{\mathrm{miss}}) = \sqrt{\left(E_{\mathrm{T}}^{x} + E_{\mathrm{T}}^{\mathrm{miss}}\right)^2 - \left|\mathbf{p}_{\mathrm{T}}^{x} + \mathbf{p}_{\mathrm{T}}^{\mathrm{miss}}\right|^2}, \tag{2}$$

  where the scalar transverse energy is evaluated as $E_{\mathrm{T}} = \sqrt{m^2 + |\mathbf{p}_{\mathrm{T}}|^2}$.

- $m_{\mathrm{T}}^0$: Projected transverse mass:

$$m_{\mathrm{T}}^0 = \sqrt{\left(p_{\mathrm{T}}^{\ell_1} + p_{\mathrm{T}}^{\ell_2} + E_{\mathrm{T}}^{\mathrm{miss}}\right)^2 - \left|\mathbf{p}_{\mathrm{T}}^{\ell_1} + \mathbf{p}_{\mathrm{T}}^{\ell_2} + \mathbf{p}_{\mathrm{T}}^{\mathrm{miss}}\right|^2}. \tag{3}$$

  Because the rest mass of an electron or muon is negligible in this high-energy context ($m \approx 0$), the transverse energy of a single lepton simplifies to its transverse momentum ($E_{\mathrm{T}}^{\ell} \approx p_{\mathrm{T}}^{\ell}$). The key difference between the standard di-lepton transverse mass ($m_{\mathrm{T}}$) and the projected transverse mass ($m_{\mathrm{T}}^0$) lies in how the scalar energy sum is constructed. The standard $m_{\mathrm{T}}$ treats the two leptons as a single composite system, whereas $m_{\mathrm{T}}^0$ treats the two leptons as separate massless entities and directly sums their individual scalar transverse momenta.

- $(\Delta R_{\ell\ell}, \Delta\eta_{\ell\ell}, m_{\ell\ell}, p_{\mathrm{T}}^{\ell\ell})$: Angular distance $\Delta R$, pseudorapidity difference, invariant mass, and transverse momentum of the di-lepton system.

- $(\Delta R_{jj}, \Delta y_{jj}, m_{jj}, \Delta\phi_{jj})$: Angular distance $\Delta R$, rapidity difference, invariant mass, and azimuthal angle difference of the di-jet system.

- $\dfrac{p_{\mathrm{T}}^{\ell_1} p_{\mathrm{T}}^{\ell_2}}{p_{\mathrm{T}}^{j_1} p_{\mathrm{T}}^{j_2}}$: Ratio of the scalar transverse momenta products of the leptons to the jets.

- $\min(\Delta R_{\ell,j})$: Minimum angular distance $\Delta R$ between any of the signal leptons and jets.

## 3.2 Dataset Construction and Validation Strategy

To ensure the robustness of the neural network models and prevent over-training, a 5-fold cross-validation strategy is employed, which is used in the ATLAS analysis. The total dataset is partitioned into training, validation, and test sets with a fixed ratio of 3:1:1.

The assignment of each event to a specific subset is determined by its event number. For a given fold index $i_{\text{fold}} \in \{0, 1, 2, 3, 4\}$, the splitting rule is defined as follows:

- **Test Set**: Events satisfying $(\text{Event Number} - i_{\text{fold}}) \pmod 5 = 0$.
- **Validation Set**: Events satisfying $(\text{Event Number} - i_{\text{fold}} + 1) \pmod 5 = 0$.
- **Training Set**: All remaining events that do not fall into the test or validation sets.

The training processes are divided into two distinct stages: a primary inclusive network ($\text{DNN}_{W^{\pm}W^{\pm}}$) designed to separate the EW signal from all backgrounds, and secondary polarization-specific networks ($\text{DNN}_{\text{pol}}$) trained exclusively on the EW samples to distinguish specific polarization states.

Table 10 details the exact physics processes used for these neural networks and their corresponding class labels (Signal or Background) in the reference thesis [7]. For the polarization networks, the mixed states are grouped such that "LX" denotes the sum of LL and TL states, while "TX" denotes the sum of TL and TT states.

Each fold is used to train an individual Dense-NN, resulting in an ensemble of five distinct networks. Crucially, this 5-fold application strategy is essential to prevent evaluation bias when predicting the final scores.

In the reference thesis [7], the author states that "predictions obtained using testing datasets can be assumed to be identically distributed as the predictions on measured data" and that "the prediction is calculated using the DNN where the respective events were part of the testing dataset." However, the thesis omits a direct procedural description of how to handle the experimental data, which possesses no true labels and never participates in the training, validation, or testing splits.

To bridge this methodological gap and ensure a statistically rigorous evaluation, we *infer* that the experimental data must be processed using the identical splitting rule applied to the Monte Carlo samples. Specifically, for any given event, the modulo operation $(\text{Event Number}) \pmod 5 = i_{\text{fold}}$ dictates which specific network in the ensemble ($\text{DNN}_i$) is exclusively used to score it. This method guarantees that the evaluation of the experimental data mirrors the independent evaluation of the respective MC test sets, thereby preserving the identically distributed assumption.

Table 10: Process usage and assigned labels across the different DNN training stages based on the reference thesis [7]. Labels are denoted as S = Signal, B = Background, and - = Not Used. The LX category represents the sum of the LL and TL states, whereas TX represents the sum of the TL and TT states.

| Physics Process | Gen. Events | $\mathrm{DNN}_{W^\pm W^\pm}$ | $\mathrm{DNN}_{\mathrm{pol}}$ (LL vs TX) | $\mathrm{DNN}_{\mathrm{pol}}$ (LX vs TT) |
|---|---|---|---|---|
| $W^\pm W^\pm jj$-EW LL (WW-cmf) | 215,008 | - | S | S |
| $W^\pm W^\pm jj$-EW TL (WW-cmf) | 170,387 | - | B | S |
| $W^\pm W^\pm jj$-EW TT (WW-cmf) | 456,456 | - | B | B |
| $W^\pm W^\pm jj$-EW LL (pp-cmf) | 19,673 | - | - | - |
| $W^\pm W^\pm jj$-EW TL (pp-cmf) | 21,471 | - | - | - |
| $W^\pm W^\pm jj$-EW TT (pp-cmf) | 46,320 | - | - | - |
| $W^\pm W^\pm jj$-EW (Unpol.) | 201,488 | S | - | - |
| $W^\pm W^\pm jj$-QCD | 25,375 | B | - | - |
| $W^\pm W^\pm jj$-Int. | 79,805 | B | - | - |
| $W^\pm Z$-EW | 8,273 | B | - | - |
| $W^\pm Z$-QCD (Comb.) | 49,999 | B | - | - |
| Top quark processes | 2,192 | B | - | - |
| $ZZ$ production | 2,880 | B | - | - |
| Charge-flip | 8,231 | B | - | - |

## 3.3 Kinematic Distributions

To illustrate the separation power of the input features, we present the normalized distributions of several key kinematic variables. The comparisons are divided into two categories, reflecting the two-stage strategy of the Dense-NN training: (1) separating the $W^{\pm}W^{\pm}jj$-EW signal from various background processes, and (2) distinguishing between the different polarization states of the $W^{\pm}W^{\pm}jj$ electroweak production.

### 3.3.1 Signal vs Background Discrimination

In the first stage of the DNN, the goal is to isolate the inclusive $W^{\pm}W^{\pm}jj$-EW signal from the dominant backgrounds, which include $W^{\pm}W^{\pm}jj$-QCD, $W^{\pm}Zjj$-EW, and $W^{\pm}Zjj$-QCD.

Figure 3 shows the distributions of six features. The VBS signature is characterized by two forward jets with a large invariant mass ($m_{jj}$) and a large pseudorapidity gap ($|\Delta\eta_{jj}|$), making them powerful discriminants against QCD-induced processes. Furthermore, the Zeppenfeld variable ($Z^{\star}_{\ell_1}$) quantifies the centrality of the lepton relative to the tagging jets; signal leptons tend to fall between the two forward jets. General kinematics, such as the leading jet transverse momentum ($p_{\mathrm{T}}^{j_1}$), di-lepton invariant mass ($m_{\ell\ell}$), and missing transverse momentum ($E_{\mathrm{T}}^{\mathrm{miss}}$), provide additional discrimination power against the $WZ$ and QCD backgrounds.

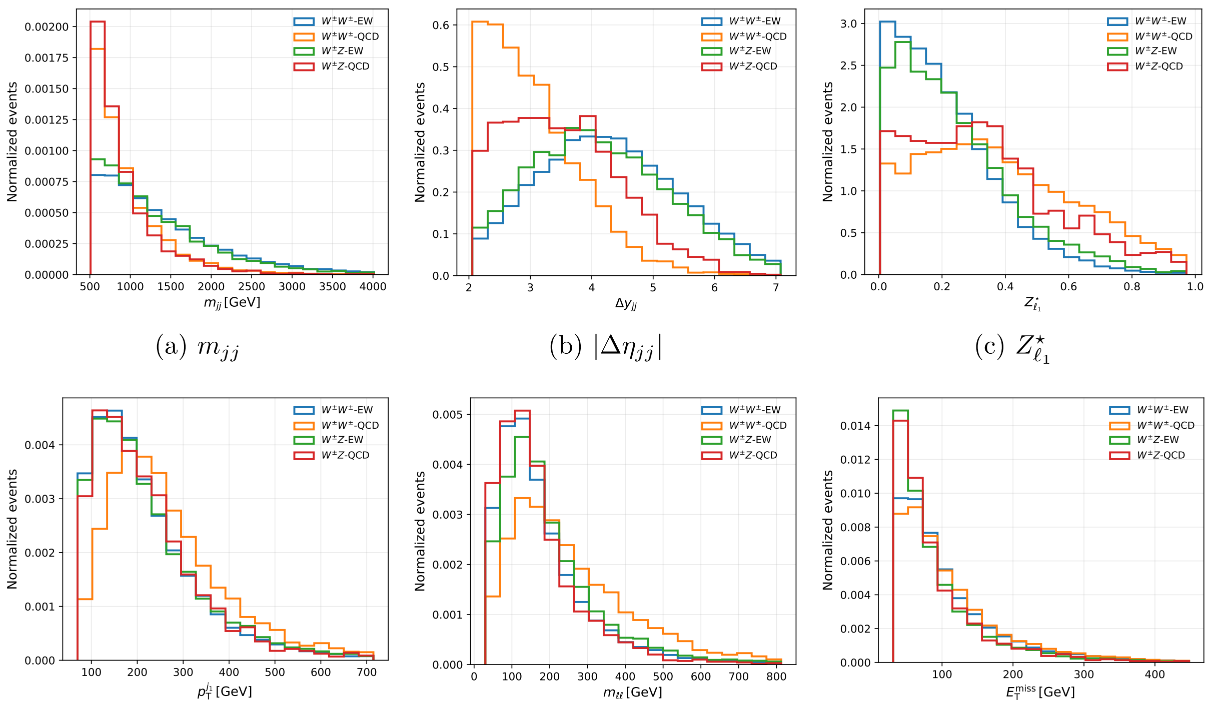
### 3.3.2 Polarization Discrimination

The second stage of the DNN focuses entirely on the $W^{\pm}W^{\pm}jj$-EW process, aiming to separate the longitudinally polarized state (LL) from the transverse states (LT and TT).

Figure 4 illustrates six variables. The $W_{\mathrm{L}}$ bosons tend to emit leptons more centrally and with different spin correlations compared to $W_{\mathrm{T}}$ bosons, leading to distinct shapes in the angular separations $|\Delta\eta_{\ell\ell}|$ and $\Delta\phi_{\ell\ell}$, as well as the leading lepton transverse momentum ($p_{\mathrm{T}}^{\ell_1}$). The transverse mass of the di-lepton system ($m_{\mathrm{T}}$) and the projected transverse mass ($m_{\mathrm{T}}^{0}$) are also sensitive to the different kinematic boundaries of the polarization states. Additionally, the azimuthal angle difference between the jets ($\Delta\phi_{jj}$) carries information about the intermediate vector boson interference in the VBS process.

## 3.4 Neural Network Architecture

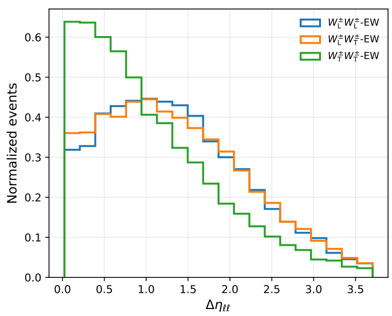
The Dense-NN architecture utilized in the analysis is presented in Figure 5.



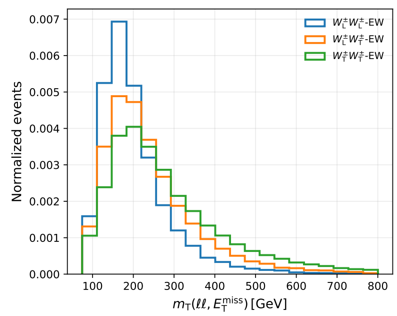

(a) $m_{jj}$ (b) $|\Delta\eta_{jj}|$ (c) $Z^{\star}_{\ell_1}$

(d) $p_{\mathrm{T}}^{j_1}$ (e) $m_{\ell\ell}$ (f) $E_{\mathrm{T}}^{\mathrm{miss}}$

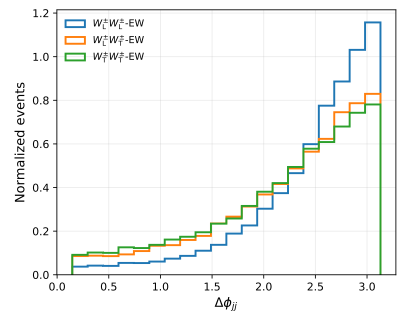
Figure 3: Normalized distributions of six key kinematic variables for the inclusive $W^\pm W^\pm jj$-EW signal and the $W^\pm W^\pm jj$-QCD, $W^\pm Zjj$-EW, and $W^\pm Zjj$-QCD backgrounds.



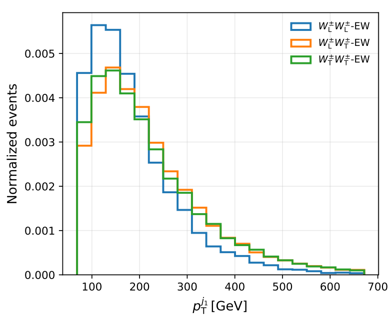

(a) $|\Delta\eta_{\ell\ell}|$



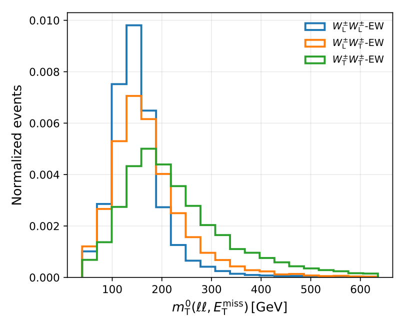

(b) $m_{\mathrm{T}}$



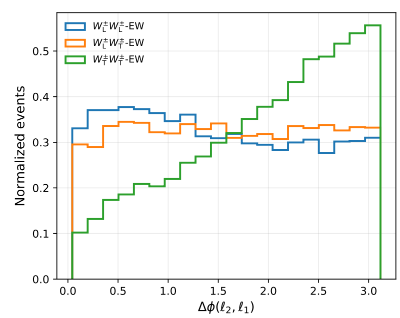

(c) $\Delta\phi_{jj}$




(d) $p_{\mathrm{T}}^{\ell_1}$




(e) $m_{\mathrm{T}}^{0}$




(f) $\Delta\phi_{\ell\ell}$

Figure 4: Normalized distributions of six polarization-sensitive variables for the LL, LT, and TT polarization states of the $W^{\pm}W^{\pm}jj$-EW process.

The network begins with an input normalization layer. This layer applies the scaling to the input features first. Then, normalizing the resulting distribution to have a mean of 0 and a variance of 1.

Following the normalization layer, the architecture relies on a core block consisting of a fully-connected (dense) layer, a Batch Normalization layer, a Swish [14] activation function, and a Dropout layer for regularization. This composite block is repeated $L$ times to form the deep hidden layers of the network.

Finally, the network terminates with an output dense layer.

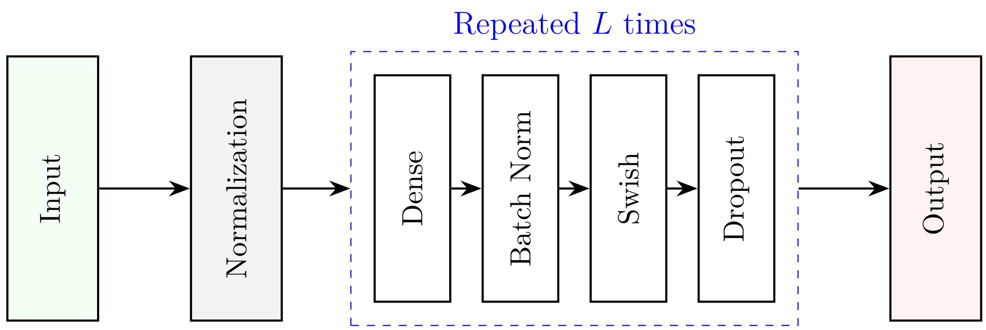



Figure 5: Schematic representation of the feed-forward neural network architecture. The sequence begins with an input normalization layer. The core architecture consists of $L$ repeated hidden layers, each comprising a dense layer with a constant number of neurons, followed by batch normalization, a Swish activation function, and a dropout layer for regularization. The final dense layer provides the binary classification output.

The Swish activation function utilized in the hidden blocks is defined as:

$$\mathrm{swish}_\beta(x) = \frac{x}{1 + \mathrm{e}^{-\beta x}} \tag{4}$$

where $\beta$ is a learnable parameter or a fixed constant. Unlike ReLU, Swish is smooth and non-monotonic, characteristics that have been shown to improve gradient propagation and performance in deep networks.

## 3.5 Implementation and Preliminary Validation

The neural network architecture, data pipeline, and training loop are implemented using `PyTorch 2.5.1` [15]. To verify the computational setup, we conducted a preliminary validation test on both the primary inclusive task $\mathrm{DNN}_{W^\pm W^\pm}$ and the secondary polarization task $\mathrm{DNN}_{\mathrm{pol}}$.

For this initial test, we utilized the complete set of input features described in Section 3.1. The dataset comprised approximately 40,000 events for the $\mathrm{DNN}_{W^\pm W^\pm}$ task and

73,000 events for the $\mathrm{DNN}_{\mathrm{pol}}$ task. The models were trained for 10 epochs using the Adam optimizer with a learning rate of $10^{-3}$ and a batch size of 256. This configuration verified the 5-fold cross-validation as well as the Dense-NN forward and backward propagation.

Figure 6 illustrates the evolution of the loss and the Area Under the ROC Curve (AUC) across the initial training epochs. The quantitative performance of the models across the 5-fold ensemble is summarized in Table 11. The negligible discrepancy between the validation and test AUC scores indicates that the data splitting strategy effectively prevents data leakage and over-training, even at this early stage.

Table 11: Preliminary performance of the Dense-NN models after 10 epochs of training. The values represent the mean AUC score and its standard deviation across the 5 independent folds.

| Classification Task | Validation AUC | Test AUC |
|---|---|---|
| $\mathrm{DNN}_{W^{\pm}W^{\pm}}$ | $0.786 \pm 0.078$ | $0.788 \pm 0.076$ |
| $\mathrm{DNN}_{\mathrm{pol}}$ (LL vs. TX) | $0.948 \pm 0.021$ | $0.947 \pm 0.024$ |

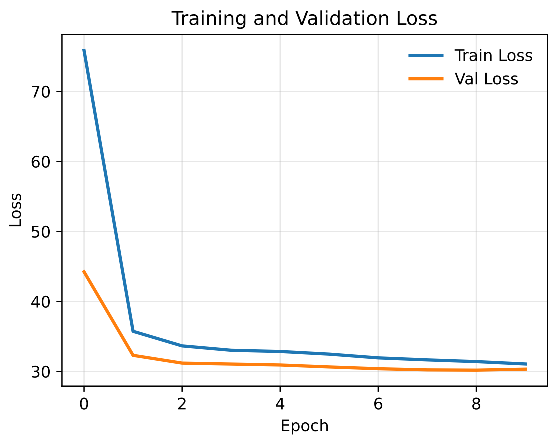
As anticipated for a preliminary test, the training curves indicate that the models have not yet converged. The next steps will involve generating larger training samples, increasing the number of epochs, and testing the optimized hyperparameters.

### 3.6 Feature Scaling, Normalization, and Full Training

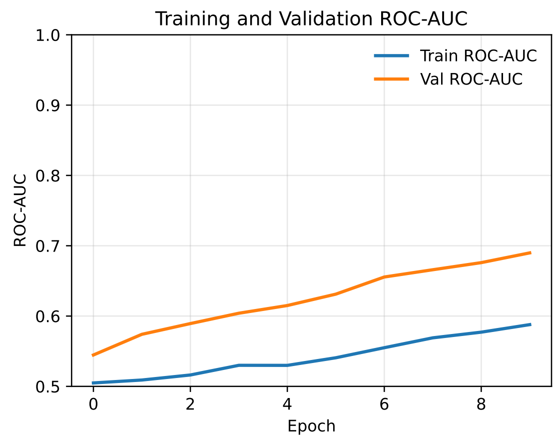
In Section 3.5, the preliminary training results for the $\mathrm{DNN}_{\mathrm{pol}}$ task yielded an unreasonably high AUC (AUC $\approx 0.948$). Upon thorough investigation, it was a result of a subtle data leakage issue during the feature normalization process. Initially, the normalization procedure (calculating the mean and standard deviation) was applied to each physics process separately before merging them into a unified dataset. This encoded class-specific properties into the scaled variables, introducing a weak label leakage that allowed the network to artificially distinguish the processes.

This issue has been resolved. The scaling function is now strictly fitted on the completely merged *training* dataset. The derived parameters are subsequently applied to transform the training, validation, and test sets.

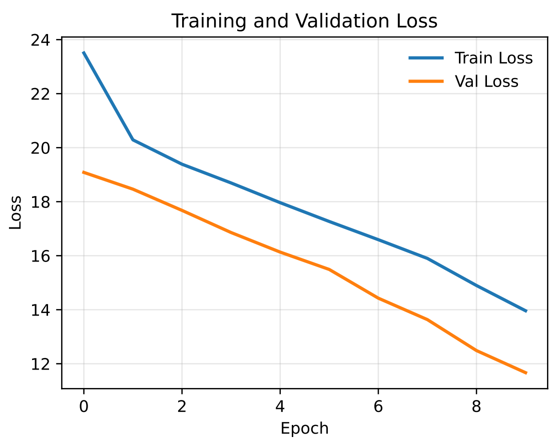
With the corrected data pipeline, we executed a full-scale training utilizing the complete set of input features described in Section 3.1. The dataset size was expanded to approximately 400,000 events for the inclusive $\mathrm{DNN}_{W^{\pm}W^{\pm}}$ task and 740,000 events for the $\mathrm{DNN}_{\mathrm{pol}}$ task.



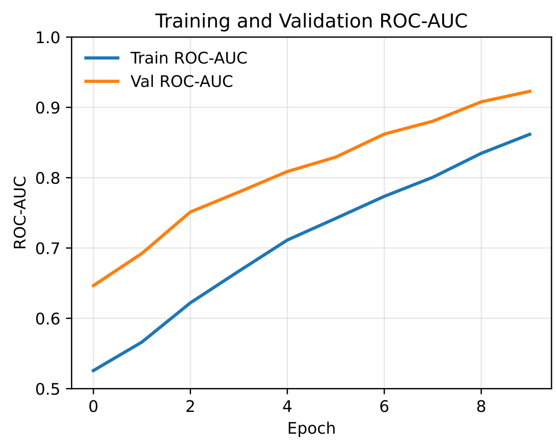

(a) Loss ($\mathrm{DNN}_{W^\pm W^\pm}$)




(b) AUC ($\mathrm{DNN}_{W^\pm W^\pm}$)




(c) Loss ($\mathrm{DNN}_{\mathrm{pol}}$)




(d) AUC ($\mathrm{DNN}_{\mathrm{pol}}$)

Figure 6: Preliminary training and validation curves (Loss and AUC) over 10 epochs for the inclusive and polarization DNN tasks.

The networks were trained using the Adam optimizer with a learning rate of $10^{-3}$ and a batch size of 256. To prevent overfitting during this extended run, an early stopping mechanism was implemented. The validation loss was monitored, and the training was terminated if no improvement was observed for 20 consecutive epochs. Table 12 summarizes the hyperparameter settings.

Table 12: Hyperparameters used in Dense-NN training.

| Parameter | Value |
|---|---|
| Number of hidden layers | 4 |
| Number of nodes per layer | 128 |
| Dropout rate | 0.3 |
| Batch size | 256 |
| Learning rate | 0.001 |
| Early stopping patience | 20 |

Figure 7 illustrates the evolution of the loss and the AUC across the training epochs. The quantitative performance of the fully trained models across the 5-fold ensemble is summarized in Table 13.

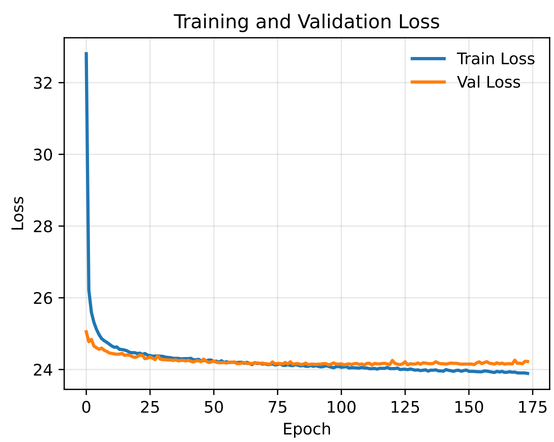
Table 13: Final performance of the Dense-NN models across the 5 independent folds after full training with early stopping. The table presents the mean AUC scores and their standard deviations. The corrected normalization strategy yields mathematically robust and realistic separation power.

| Classification Task | Validation AUC | Test AUC |
|---|---|---|
| $\mathrm{DNN}_{W^\pm W^\pm}$ (EW vs. Background) | $0.806 \pm 0.001$ | $0.806 \pm 0.001$ |
| $\mathrm{DNN}_{\mathrm{pol}}$ (LL vs. TX) | $0.839 \pm 0.001$ | $0.839 \pm 0.001$ |

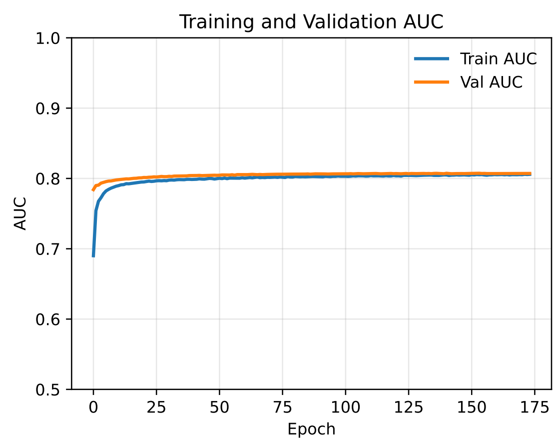
The revised test AUC for the polarization task represents a much more realistic separation power. Furthermore, the small variance between the folds and the negligible discrepancy between the validation and test AUC scores confirm that the data splitting and normalization strategy effectively prevents overtraining.

## 3.7 DNN Output Score Distributions

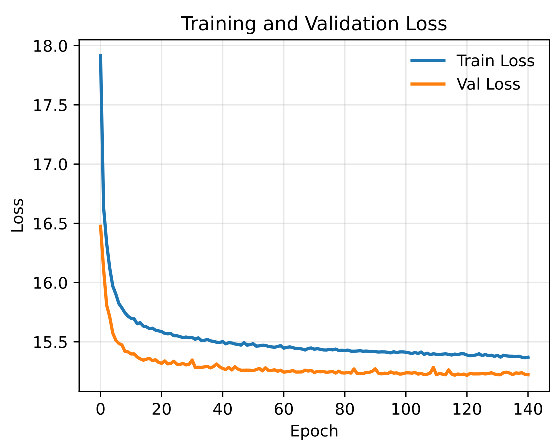
To visually assess the discrimination power of the trained networks, we evaluate the DNN output score distributions for both the inclusive and polarization-specific classification



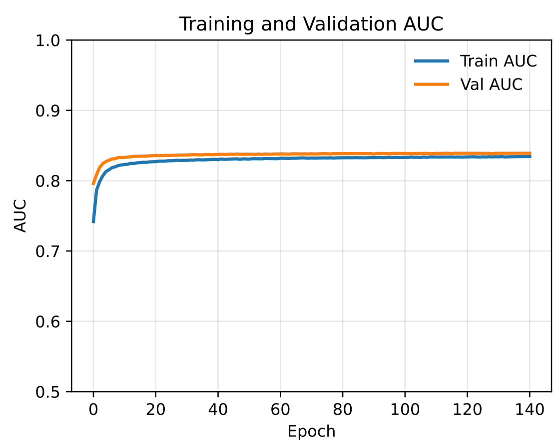

(a) Loss ($\mathrm{DNN}_{W^\pm W^\pm}$)




(b) AUC ($\mathrm{DNN}_{W^\pm W^\pm}$)



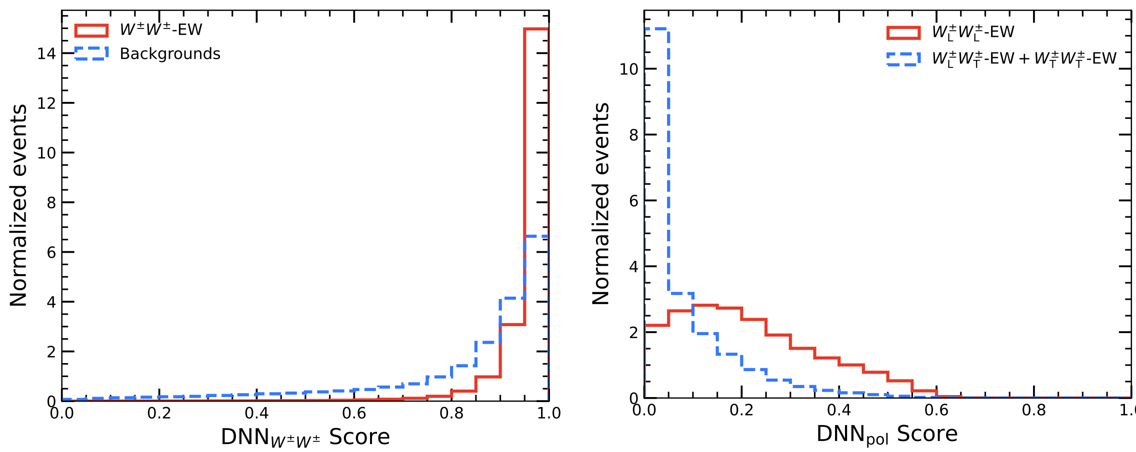

(c) Loss ($\mathrm{DNN}_{\mathrm{pol}}$)




(d) AUC ($\mathrm{DNN}_{\mathrm{pol}}$)

Figure 7: Full training and validation curves (Loss and AUC) for the inclusive and polarization Dense-NN tasks. The early stopping mechanism successfully prevents overtraining.

tasks. Figure 8 presents the normalized predicted probability scores evaluated on the test datasets.




(a) $\mathrm{DNN}_{W^{\pm}W^{\pm}}$ Score Distribution

(b) $\mathrm{DNN}_{\mathrm{pol}}$ Score Distribution

Figure 8: Distributions of the DNN output scores for the test datasets. (a) The inclusive $\mathrm{DNN}_{W^{\pm}W^{\pm}}$ separating the EW signal from backgrounds. (b) The $\mathrm{DNN}_{\mathrm{pol}}$ distinguishes the LL signal from the TX states.

In an idealized binary classification scenario, the output scores would exhibit distinct peaks at 1.0 for the signal and 0.0 for the background. However, as observed in Figure 8, the distributions display a profile with noticeable overlapping regions.

Upon preliminary investigation, this behavior is attributed to the incorporation of physical event weights during the training process. However, we observed a discrepancy between our current score distributions and the reference plots presented in the thesis [7].

## 3.8 Training Weight Strategy

To properly account for the different production cross-sections and the imbalanced MC dataset sizes, we evaluated three distinct sample weighting strategies for the NN training loss function:

1. **Physical Yield Weighting:** Each event is assigned a weight corresponding to its expected physical yield in the SR.

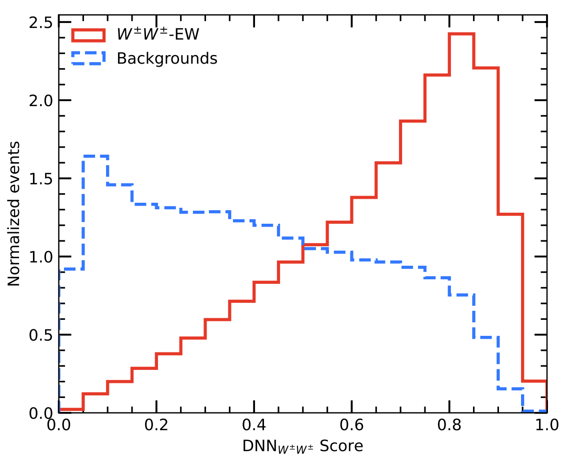
2. **Inverse Generation Weighting:** Each event is weighted simply by the inverse of the total number of generated events for its respective process ($w \propto 1/N_{\mathrm{gen}}$).

3. **Hybrid Weighting:** A combination of the physical yield and inverse event count methods.

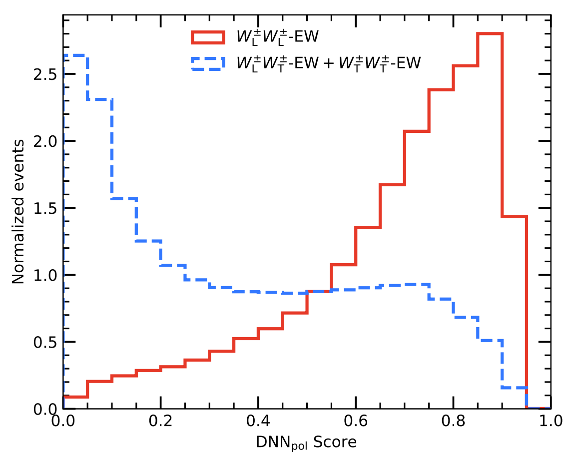
The pure "Physical Yield" strategy reflects the true kinematic composition and imbalance among all physical processes, closely mimicking the exact conditions of real collision data. However, if used alone, it fails to account for the imbalance within the generated MC samples themselves. This is why the "Inverse Generation" and "Hybrid" approaches were introduced and evaluated.

Furthermore, a crucial **class-balancing** step is enforced. The event weights are globally normalized such that the sum of the weights in the signal category exactly equals the sum of the weights in the background category ($\sum w_{\text{sig}} = \sum w_{\text{bkg}}$). This global scaling preserves the relative physical ratios of the internal sub-processes while preventing the neural network from trivially predicting the majority class to minimize the loss.

The Hybrid weighting strategy, combined with global class balancing, provides the most reasonable learning strategy. Thus, we presented the training results of this strategy. Figure 9 illustrates the resulting Dense-NN output evaluated on the test datasets utilizing this optimized weighting scheme. The distributions satisfy our expectation that they exhibit peaks near 1 for the signal and 0 for the background.




(a) $\text{DNN}_{W^{\pm}W^{\pm}}$ Score Distribution




(b) $\text{DNN}_{\text{pol}}$ Score Distribution

Figure 9: Distributions of the DNN output scores for the test datasets using the optimized hybrid weighting strategy. (a) The inclusive $\text{DNN}_{W^{\pm}W^{\pm}}$ separating the EW signal from backgrounds. (b) The $\text{DNN}_{\text{pol}}$ distinguishes the LL signal from the TX states.

# 4 Statistical framework

This section outlines the statistical framework utilized to extract the final physics results. The ultimate goal of this analysis is to evaluate the Discovery Significance $Z_0$ and the Upper Limit constraints.

## 4.1 Profile Likelihood Fit

First, we need to compute the profile likelihood $\mathcal{L}$, which evaluates the goodness-of-fit between the theoretical model and the observed data across the 1D histograms. For each bin $i$, the expected number of events is defined as $m_i$, and the observed number of events is denoted as $n_i$.

The expectation $m_i$ consists of the signal and background contributions, parameterized by the signal strength $\mu$, and is modified by systematic uncertainties. These uncertainties are modeled using nuisance parameters (NPs), denoted as $\boldsymbol{\theta}$ for systematics, and $\boldsymbol{\gamma}$ for Monte Carlo (MC) statistical uncertainties:

$$m_i(\mu, \boldsymbol{\theta}, \boldsymbol{\gamma}) = \gamma_i \left(\mu\, s_i(\boldsymbol{\theta}) + b_i(\boldsymbol{\theta})\right), \tag{5}$$

where $s_i$ and $b_i$ are the nominal number of predicted signal and background events in bin $i$, respectively.

The complete likelihood function $\mathcal{L}$ is constructed as the product of Poisson probabilities for all bins, multiplied by constraint terms for the nuisance parameters:

$$\mathcal{L}(\mu, \boldsymbol{\theta}, \boldsymbol{\gamma}) = \prod_{i=1}^{N_{\text{bins}}} \text{Pois}\left(n_i | m_i\right) \text{Pois}\left(\frac{1}{(\delta_i^{\text{stat}})^2} \middle| \frac{\gamma_i}{(\delta_i^{\text{stat}})^2}\right) \prod_{j=1}^{N_{\text{nuis}}} f_{\text{Gauss}}(\theta_j). \tag{6}$$

Here, the Poisson probability is defined as $\text{Pois}(n_i|m_i) = \frac{m_i^{n_i}}{n_i!} \mathrm{e}^{-m_i}$. The auxiliary functions $f_{\text{Gauss}}(\theta_j)$ are standard normal distributions constraining the systematic NPs $\boldsymbol{\theta}$.

### 4.1.1 Test Statistic and Profile Likelihood Ratio

To test specific hypotheses, the profile likelihood ratio $\lambda(\mu)$ is defined as:

$$\lambda(\mu) = \frac{\mathcal{L}(\mu, \hat{\hat{\boldsymbol{\theta}}}, \hat{\hat{\boldsymbol{\gamma}}})}{\mathcal{L}(\hat{\mu}, \hat{\boldsymbol{\theta}}, \hat{\boldsymbol{\gamma}})}. \tag{7}$$

In the denominator, $\hat{\mu}$, $\hat{\boldsymbol{\theta}}$, and $\hat{\boldsymbol{\gamma}}$ are the unconditional maximum likelihood estimators that globally maximize $\mathcal{L}$ across the entire parameter space. In the numerator, $\hat{\hat{\boldsymbol{\theta}}}$ and $\hat{\hat{\boldsymbol{\gamma}}}$ are the conditional maximum likelihood estimators that maximize $\mathcal{L}$ for a specified, fixed value of $\mu$.

Based on $\lambda(\mu)$, the test statistic $q_\mu$ is constructed:

$$q_\mu = -2 \ln \lambda(\mu). \tag{8}$$

### 4.1.2 Discovery Significance and Upper Limits

The presence of a new signal is tested by rejecting the background-only hypothesis ($\mu = 0$). The corresponding test statistic is $q_0 = -2 \ln \lambda(0)$. For a sufficiently large dataset, the discovery significance $Z_0$ can be approximated as:

$$Z_0 \approx \sqrt{q_0}. \tag{9}$$

In the absence of a significant excess, the analysis sets a 95% Confidence Level (CL) upper limit on the signal strength. This is derived using the $\mathrm{CL}_s$ method, which also utilizes the test statistic $q_\mu$.

## 4.2 The histogram binning

In this analysis, the outputs of the Dense-NNs serve as the direct inputs for the profile likelihood fit. The outputs can be interpreted as classification tuple $(\mathrm{DNN}_{\mathrm{pol}}, \mathrm{DNN}_{W^\pm W^\pm})$ on a two-dimensional score plane:

- Y-axis ($\mathrm{DNN}_{W^\pm W^\pm}$): The inclusive network score, which separates the pure EW $W^\pm W^\pm jj$ signal from reducible and irreducible backgrounds (e.g., $WZ$, QCD $WW$).
- X-axis ($\mathrm{DNN}_{\mathrm{pol}}$): The polarization-specific network score, which evaluates the EW events to distinguish the targeted polarization state (e.g., LL signal) from the other polarization states (e.g., TX backgrounds).

This study uses a rebinning algorithm to maximize the significance and reduce the computation time. This procedure slices the 2D plane into a sequence of 1D bins, requiring a delicate balance between preserving the Dense-NN output shape information and avoiding statistical fluctuations. The algorithm is listed below:

- **Region Slicing:** The Y-axis ($\mathrm{DNN}_{W^\pm W^\pm}$) is first divided into discrete purity intervals. Each interval is then evaluated along the X-axis ($\mathrm{DNN}_{\mathrm{pol}}$) and flattened into consecutive 1D strips.

- **Statistical Stability:** To prevent statistical uncertainties, the bin edges of each histogram are merged until there are at least ten events originating from background or transverse polarized contributions in each bin.

- **Significance Maximization:** Adjacent bins are merged if keeping them separate would not improve the significance.

Table 14 is the final binning from the thesis.

Table 14: The optimized splits in the $\text{DNN}_{W^\pm W^\pm}$ classification and the corresponding bin edges in polarization classification.

| Polarization | Split in $\text{DNN}_{W^\pm W^\pm}$ | Binning in polarization classification |
|---|---|---|
| LL in $WW$-cmf | $[0.0, 0.2, 0.6, 1.0]$ | $[0.0, 0.4, 0.7, 1.0]$<br>$[0.0, 0.5, 0.8, 1.0]$<br>$[0.0, 0.4, 0.6, 0.7, 0.8, 1.0]$ |
| LX in $WW$-cmf | $[0.0, 0.3, 0.7, 1.0]$ | $[0.0, 0.3, 0.7, 1.0]$<br>$[0.0, 0.3, 0.5, 0.6, 0.7, 0.8, 1.0]$<br>$[0.0, 0.15, 0.35, 0.45, 0.55, 0.7, 1.0]$ |
| LL in pp-cmf | $[0.0, 0.15, 0.5, 1.0]$ | $[0.0, 0.55, 0.8, 1.0]$<br>$[0.0, 0.7, 0.8, 1.0]$<br>$[0.0, 0.4, 0.6, 0.7, 0.8, 1.0]$ |
| LX in pp-cmf | $[0.0, 0.25, 0.6, 1.0]$ | $[0.0, 0.45, 0.6, 1.0]$<br>$[0.0, 0.25, 0.4, 0.55, 0.75, 1.0]$<br>$[0.0, 0.15, 0.3, 0.4, 0.5, 0.6, 0.7, 1.0]$ |

## 4.3 Expected Results

The fit model must be validated using an expected dataset. This is achieved by utilizing an **Asimov dataset**, which is a representative pseudo-dataset where the "observed" event counts in each bin are set exactly equal to the nominal theoretical predictions (signal plus background), absent any statistical fluctuations.

Fitting the Asimov dataset serves a critical purpose: it determines the theoretical sensitivity of the analysis based on the current detector acceptance, background estimations, and the discrimination power of the DNN. This yields a baseline expected discovery significance ($Z_0$) for the targeted polarization states.

Table 15 summarizes the expected significances for the LL and LX signal states as reported in the reference thesis [7].

Table 15: Expected discovery significances for the LL and LX polarization states, evaluated using the Asimov dataset. The comparison highlights the impact of defining the polarization vectors in the $WW$ center-of-mass frame versus the partonic frame.

| Reference Frame | LL Significance ($Z_0$) | LX Significance ($Z_0$) |
|---|---|---|
| WW-cmf | $1.512\,\sigma$ | $4.557\,\sigma$ |
| pp-cmf | $0.937\,\sigma$ | $3.975\,\sigma$ |

As shown in the table, defining the polarization in the $WW$ center-of-mass frame yields notably higher expected sensitivities for both the LL and LX states compared to the initial-state partonic frame.

## 4.4 Results of the Statistical Fit

Using the profile likelihood fit framework and the unrolled 1D binning configuration described above, we evaluate the expected discovery significance $Z_0$ and the expected 95% CL upper limit on the signal strength parameter $\mu$ for both the LL mode and the LX mode. The fit is performed using the Asimov dataset constructed from our own simulated Monte Carlo samples in the $WW$ center-of-mass frame.

Table 16 summarizes our obtained expected discovery significances and expected 95% CL upper limits on the signal strength parameters, alongside the reference results from the thesis [7] for comparison.

Table 16: Expected discovery significances ($Z_0$) and expected 95% CL upper limits on the signal strength parameters ($\mu$) for the LL and LX polarization states in the $WW$ center-of-mass frame. The reference values from the thesis are provided for comparison.

| Signal State | Expected Significance ($Z_0$) | | Expected 95% CL Upper Limit | |
|---|---|---|---|---|
| | Thesis | Our Result | Thesis | Our Result |
| LL ($\mu_{\mathrm{LL}}$) | $1.512\,\sigma$ | $1.600\,\sigma$ | 2.273 | 1.255 |
| LX ($\mu_{\mathrm{LX}}$) | $4.557\,\sigma$ | $4.852\,\sigma$ | — | 0.356 |

As shown in Table 16, despite utilizing unoptimized Dense-NNs and slightly varied input features, our simulation successfully reproduces the overall statistical trend. The expected significances are in agreement with the reference values.

However, a discrepancy is observed in the expected 95% CL upper limit for the LL state, where our result is approximately half of the reference value. This deviation directly highlights the impact of Nuisance Parameters (NPs). Our current fit considers only statistical uncertainties. In a full fit, systematic uncertainties flatten and widen the profile likelihood curve. Consequently, the absence of these systematic constraints results in a tighter upper limit.

# 5 Some research directions

We propose some research directions to further enhance the discovery significance and methodological robustness of the longitudinal $W$ boson scattering study.

## 5.1 Enhance the discovery significance

The current analysis relies on a baseline Dense-NN, which flattens kinematic features and may lose underlying spatial correlations. We propose to utilize advanced architectures to boost classification accuracy and then enhance the discovery significance:

- **Advanced Architectures:** Replace the Dense-NN with CNN, **ParticleNet** (Graph Neural Networks, GNNs) [16], or **Particle Transformer** (ParT) [17] to better capture the complex VBS topologies.
- **Multi-Task Learning (MTL):** Instead of training separate classifiers for inclusive $WW$ signal and polarization states, implement a single MTL framework with a shared representation space and multiple classification heads. This can yield better generalization and higher separation power.
- **Multi-Class classification:** Instead of utilizing the step separation strategy, implement a network to distinguish LL, LT, TT, and other background classes directly.

## 5.2 Unsupervised / Self-supervised learning

Current supervised training heavily relies on MC simulations, which introduce theoretical uncertainties and potential biases when applied to measured data.

- **Weakly Supervised Learning & Topic Modeling:** The Classification Without Labels (CWoLa) paradigm [18] successfully operates on two mixed data regions for binary classification. We can extend this to multi-class scenarios using Topic Modeling [19]. This approach treats different mixed sample regions as “documents” and the underlying physics processes (e.g., pure LL, LT, TT states) as “topics,” enabling the extraction of pure kinematic distributions directly from collision data.

- **Self-Supervised Contrastive Learning:** Implementing self-supervised frameworks, such as JetCLR [20], allows the neural network to learn robust, physics-invariant representations without requiring explicit MC labels. By applying physical data augmentations (e.g., azimuthal rotations, longitudinal boosts, or momentum smearing), the model maps physically similar events closer together in a latent space. This latent representation can subsequently be evaluated or fine-tuned using minimal MC labels via linear probing.

- **Objective and Impact:** The ultimate goal is to exploit these data-driven techniques to construct optimal kinematic observables and isolate the polarization states with minimal dependence on pure MC labels.

### 5.3 Fully Hadronic Channel

While the baseline analysis focuses on the clean same-sign fully leptonic channel ($\ell^\pm\nu\ell^\pm\nu$), the fully hadronic decay mode ($WW \to qqqq$) represents a compelling yet highly challenging frontier for VBS polarization studies. To date, no experimental results for VBS exist in this channel.

## 6 Fully Hadronic Channel on VBS analysis

The study of Vector Boson Scattering (VBS) in the fully hadronic decay channel ($VV \to 4q$ and $ZV \to 2\nu 2q$, where $V = W, Z$) provides a critical test of the Standard Model due to its large branching fraction. In this section, we summarize the analysis status, review the literature on fully hadronic VBS, and discuss potential phenomenological research directions.

### 6.1 Summary of literature review

While polarized vector boson scattering has been studied in leptonic channels, there are currently no experimental results for polarized VBS (such as $W_\mathrm{L}W_\mathrm{L}$) in the fully hadronic channel. Even for unpolarized Standard Model VBS, no public experimental observations in the fully hadronic channel have been reported to date. This is due to the lack of distinct resonant peaks and the presence of an overwhelming QCD multijet background. This absence of fully hadronic SM VBS measurements is highlighted in ATLAS conference (2024) reviews [21] and a recent (2026) VBS review article [22], which omit the fully hadronic final state entirely. Currently, the only detailed study of the unpolarized Standard Model VBS in the fully hadronic final state is documented in the Ph.D. thesis of Miao Hu [23], which is discussed in detail in Section 6.2.

In contrast to the Standard Model VBS measurements, the only public experimental results in the fully hadronic final state at the LHC are searches for heavy resonances decaying to boosted vector boson pairs ($VV \to qqqq$), such as those conducted by ATLAS [24] and CMS [25]. These searches exploit highly boosted topologies where the decay products of each vector boson are collimated into a single large-radius jet. By employing advanced jet substructure techniques, these analyses successfully suppress the massive QCD multijet background.

On the phenomenological side, studies have investigated the potential of the hadronic VBS final state to probe new physics. For instance, Ref. [26] has explored the sensitivity of the LHC to the resonance in the $WW$ hadronic channel using boosted topologies.

## 6.2 Summary of the Hadronic VBS Analysis

The key aspects of the analysis in Ref. [23] are summarized below:

- **Event Categorization:** One of the vector bosons is required to decay in a hadronic channel only, and this vector boson can be boosted or resolved. The other vector bosons are considered in three decay modes: $Z \to \nu\nu$, boosted $Z \to c\bar{c}/b\bar{b}$, or boosted $V \to q'q'$. Thus, $2 \times 3 = 6$ categories are supposed to be identified. However, two all-visible resolved regions are not considered.
  - **Boosted-Invisible (B-MET):** One boosted $V \to qq$ and one invisibly decaying $Z \to \nu\nu$, characterized by high $E_{\mathrm{T}}^{\mathrm{miss}}$ and one AK8 jet.
  - **Boosted-withB (B-Btag):** Two boosted $V$ bosons with at least one decaying to heavy-flavor quarks ($Z \to b\bar{b}/c\bar{c}$).
  - **Boosted-Boosted (B-B):** Two boosted $V$ bosons decaying to light quarks, characterized by low $E_{\mathrm{T}}^{\mathrm{miss}}$ and two identified AK8 jets.
  - **Resolved-Invisible (R-MET):** One resolved $V \to qq$ and one invisibly decaying $Z \to \nu\nu$, characterized by high $E_{\mathrm{T}}^{\mathrm{miss}}$, zero AK8 jets, and multiple AK4 jets.
- **Results:** Using a simultaneous fit to the MVA templates across the signal and control regions, the combined analysis reports an expected significance of $3.9\,\sigma$ (specifically $3.88\,\sigma$) for the EW $VV$ production.

## 6.3 Possible research directions

In the experimental CMS analysis [23], the two "all-visible" categories are discarded. This is due to trigger bottlenecks at the LHC as well as the overwhelming QCD multijet background. Both experimental searches and phenomenological VBS studies have focused almost exclusively on boosted boson topologies. Currently, there is no discussion in the literature regarding the feasibility of a fully resolved hadronic VBS final state, where all vector boson decay products are resolved into individual jets.

We propose to investigate the feasibility of the fully resolved VBS hadronic final state. The primary limitations in this channel are the complex jet combinatorics and signal-background separation. To address these challenges, we can employ advanced machine learning architectures:

- Jet Pairing and Classification with SPA-NET: In the fully resolved channel ($VVjj \to 6j$), matching the six reconstructed jets to their parent particles is a severe combinatorics problem. We can utilize Symmetry Preserving Attention Networks (SPA-NET) to perform jet pairing and event classification. By training SPA-NET to learn the underlying symmetries of the VBS decay topology, we can reconstruct the parent bosons and distinguish the VBS signal from the QCD multijet background.

  Furthermore, as discussed in Ref. [27], generalized SPA-NET models can simultaneously handle both boosted and resolved topologies, offering a unified reconstruction framework for future studies.

## 6.4 Proposed analysis workflow for the resolved channel

To implement the fully resolved hadronic VBS polarization analysis, we propose the following step-by-step workflow:

1. **Event Generation and Truth Matching:** Generate Monte Carlo samples for the signal electroweak process ($pp \to WWjj \to qqqqjj$ in polarized states $W_\mathrm{L}W_\mathrm{L}$, $W_\mathrm{L}W_\mathrm{T}$, and $W_\mathrm{T}W_\mathrm{T}$) and the dominant QCD multijet background. A truth-matching script will match reconstructed $R = 0.4$ jets to generator-level quarks to label the correct jet-parton assignments for SPA-NET training.

2. **Pre-selection cuts:** Apply basic event selection cuts to identify VBS candidate events, such as requiring at least six reconstructed jets with $p_\mathrm{T} > 20\,\mathrm{GeV}$ and $|\eta| < 5$. Reference the Ph.D. thesis of Miao Hu [23].

3. **Spa-Net training and inference:** Train Spa-Net on the truth-matched resolved samples to perform jet-parton assignment and classify electroweak signal against the QCD background. During inference, Spa-Net will output the most probable jet-parton pairing and event-level classification scores.

4. **Observable reconstruction and Neural Network scores:** Using the jet pairings predicted by Spa-Net, we reconstruct the parent $W$ bosons. We can either construct physical decay angles or utilize the output classification scores from the neural network.

5. **Statistical fit:** Use the resulting distributions to perform a profile likelihood fit to evaluate the expected discovery significance or the upper limit for the longitudinal polarization (or the $WW$ channel).

# 7 Particle Transformer

As a next step to enhance the classification performance, we explore advanced neural network architectures. First, we investigate the Particle Transformer (ParT) [17].

## 7.1 Model Architecture and Mathematical Details

The ParT architecture is designed for jet tagging. By representing the jet as a permutation-invariant particle cloud, the model avoids positional encodings. The core mathematical formulation and the structural design of the model are detailed below.

### 7.1.1 High-Level Architecture Overview

The global data flow of the ParT is illustrated in Figure 10. The model processes two parallel branches of inputs: the particle features and the pairwise physical interaction features, which are then passed through the attention blocks to produce classification scores.

The processing pipeline consists of the following components:

- **Input Embedding:** The input particle feature $X$ is mapped to $x^0 \in \mathbb{R}^{N\times d}$ via a 3-layer MLP. Simultaneously, the pairwise physical interaction feature $U_{\text{in}}$ is mapped to $U \in \mathbb{R}^{N\times N\times d'}$ using a 4-layer Pointwise 1D Convolution.

- **Particle Attention Blocks:** $x^0$ passes through $L$ blocks. Within each block, the interaction tensor $U$ acts as an attention bias in the Particle Multi-Head Attention (P-MHA) mechanism.

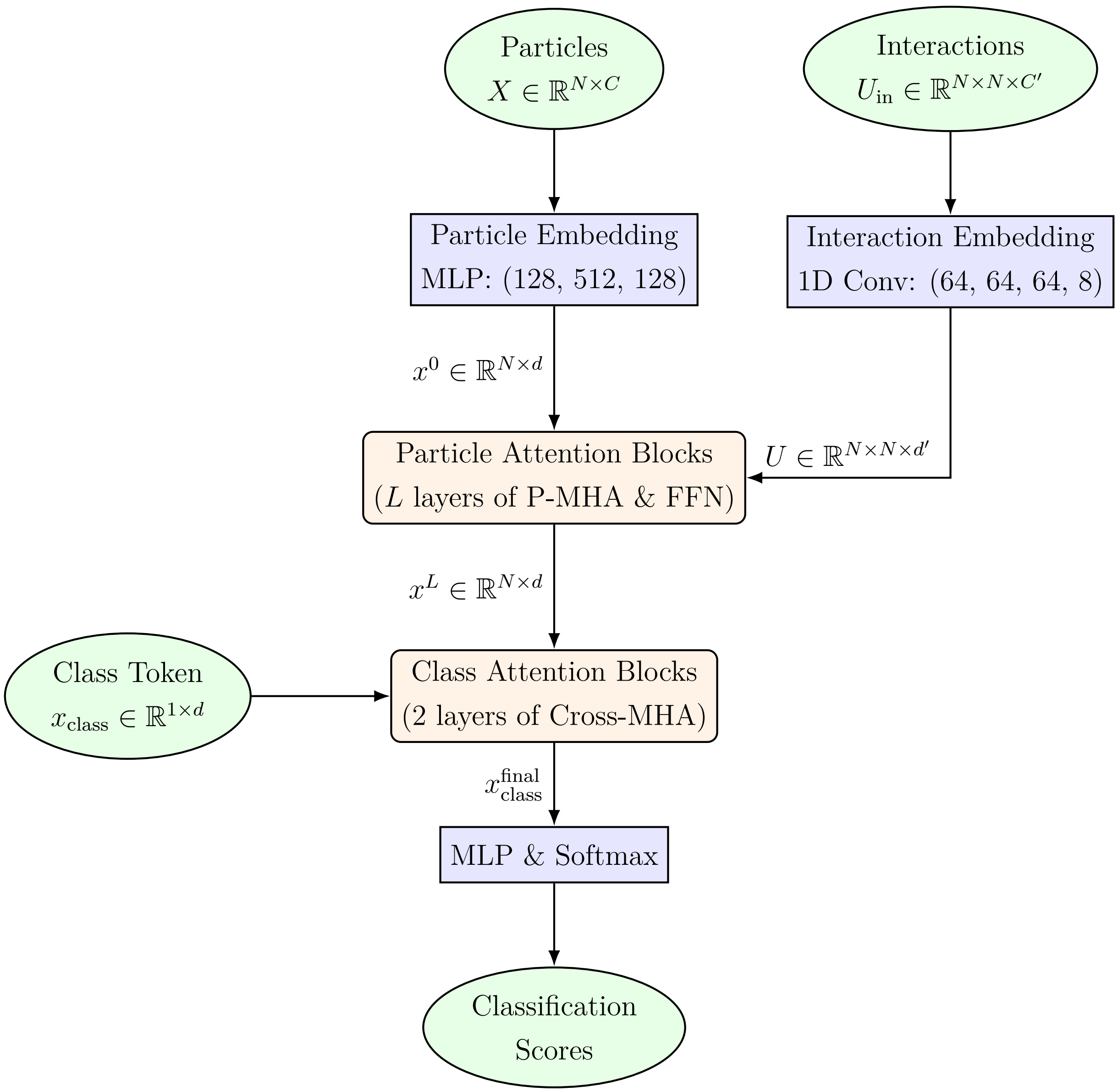



Figure 10: Data flow and high-level architecture of Particle Transformer (ParT).

- **Class Attention Blocks:** A learnable class token $x_{\text{class}}$ queries the final particle representation $x^L$ through cross-attention to aggregate event-level information (Section 7.1.5).

- **Classifier Head:** The aggregated class token is passed through a final MLP and a Softmax function to yield the classification probabilities.

### 7.1.2 Input and Attention Mechanism

Let the input sequence of $N$ particles be represented by $X \in \mathbb{R}^{N\times d}$, where $d$ is the input feature dimension. Let $U \in \mathbb{R}^{N\times N\times d'}$ represent the pairwise physical interaction features calculated between each pair of particles.

The input feature $X$ is projected into the attention space using projection matrices:

$$Q = XW_q, \quad K = XW_k, \quad V = XW_v, \tag{10}$$

where $W_q, W_k, W_v \in \mathbb{R}^{d\times d_{\text{attn}}}$. In general, the projection matrices do not need to be square matrices, but we would set $d = d_{\text{attn}}$ for simplicity.

For a multi-head attention mechanism with $H$ heads, the projection matrices are split column-wise into $H$ submatrices:

$$W = \left[W^{(1)}, W^{(2)}, \ldots, W^{(H)}\right], \tag{11}$$

where each submatrix $W^{(h)} \in \mathbb{R}^{d\times d_k}$ is rectangular, with $d_k = d_{\text{attn}}/H$ representing the feature dimension of each head.

### 7.1.3 Particle Multi-Head Attention (P-MHA)

For each head $h \in \{1, 2, \ldots, H\}$, the query, key, and value vectors are computed as:

$$Q_h = XW_q^{(h)}, \quad K_h = XW_k^{(h)}, \quad V_h = XW_v^{(h)}, \tag{12}$$

where $Q_h, K_h \in \mathbb{R}^{N\times d_k}$ and $V_h \in \mathbb{R}^{N\times d_v}$. The attention score matrix $S_h \in \mathbb{R}^{N\times N}$ for head $h$ is given by:

$$S_h = \frac{Q_h K_h^{\mathrm{T}}}{\sqrt{d_k}} + U_h \tag{13}$$

where $U_h \in \mathbb{R}^{N\times N}$ is the $h$-th channel of the projected interaction matrix $U \in \mathbb{R}^{N\times N\times d'}$. This addition imposes a constraint that the number of interaction channels $d'$ must equal the number of attention heads $H$.

The scaling factor $1/\sqrt{d_k}$ is essential to control the variance of the dot product. Assuming the elements of $q_i, k_j \in \mathbb{R}^{d_k}$ are independent random variables with mean 0 and variance

1, the variance of their dot product is $\text{Var}(q_i \cdot k_j) = d_k$. Without scaling, a large $d_k$ would result in extreme attention scores, pushing the Softmax function into the saturation region. Scaling ensures the input to the Softmax function has a unit variance:

$$\text{Var}\left(\frac{q_i \cdot k_j}{\sqrt{d_k}}\right) = 1. \tag{14}$$

The output of head $h$, $O_h \in \mathbb{R}^{N \times d_v}$, is calculated as:

$$O_h = \text{Softmax}(S_h)V_h. \tag{15}$$

Note that in general, $d_k$ is not required to equal $d_v$.

#### 7.1.4 Concatenation, Output Projection, and Residual Connection

The outputs from all $H$ heads are concatenated along the feature dimension:

$$O_{\text{concat}} = [O_1, O_2, \ldots, O_H] \in \mathbb{R}^{N \times (H \cdot d_v)}. \tag{16}$$

A final projection matrix $W_o \in \mathbb{R}^{(H \cdot d_v) \times d_{\text{out}}}$ is applied:

$$\text{Output} = O_{\text{concat}} W_o \in \mathbb{R}^{N \times d_{\text{out}}}. \tag{17}$$

To support the residual connection, we set the output dimension to match the input dimension ($d_{\text{out}} = d$), allowing the addition:

$$X_{\text{out}} = \text{LayerNorm}\left(\text{Output}\right) + X_{\text{in}}. \tag{18}$$

The constraints $d = d_{\text{attn}}$ and $d_k = d_v$ are not mathematical necessities, but rather simplifications adopted in standard implementations.

#### 7.1.5 Class Token and Information Aggregation

At the end of the particle attention blocks, a learnable global class token $x_{\text{class}} \in \mathbb{R}^{1 \times d}$ is introduced. The class token acts as a query in cross-attention to aggregate information from all particles:

$$Q_{\text{class}} = x_{\text{class}} W_q, \tag{19}$$

$$K_{\text{class}} = z W_k, \quad V_{\text{class}} = z W_v, \tag{20}$$

where $z = [x_{\text{class}}; X_{\text{particles}}] \in \mathbb{R}^{(1+N) \times d}$.

This design offers significant advantages:

- **Dynamic Aggregation:** Unlike static flat projections, the class token dynamically determines which particles to focus on based on context.
- **Permutation Invariance:** The cross-attention mechanism is permutation invariant, ensuring that the ordering of the constituents does not affect the classification result.
- **Variable Sequence Length:** Since the attention matrix size scales with the number of particles $N$, the output shape of the class token remains constant $(1 \times d)$ regardless of $N$, naturally accommodating events with varying numbers of constituents.

The final class token is extracted and fed into a multi-layer perceptron for classification.

## 7.2 Input Feature Sets

To evaluate the potential of ParT in VBS polarization studies, we test two distinct input feature sets: a high-level representation based on reconstructed physics objects and a low-level representation based on detector-level constituents.

- **High-Level Feature Set (6 features per object):** The inputs are constructed from the main reconstructed physics objects in the signal region: the 2 leptons, the 2 tagging jets, and the missing transverse energy (MET), forming a sequence of 5 objects in total. For each object, the input features consist of:
  - $p_{\mathrm{T}}$: Transverse momentum.
  - $\eta$: Pseudorapidity.
  - $\Delta\phi(x, \ell_1)$: Azimuthal angle difference relative to the leading lepton $\ell_1$.
  - One-hot encoding (3-dimensional vector): Identifies the object type as lepton, jet, or MET.

  This yields a feature matrix of shape $5 \times 6$ per event.
- **Low-Level Feature Set (5 features per constituent):** The inputs are constructed from the detector-level tracks and calorimeter towers. To maintain a fixed sequence size for the transformer, a maximum of 128 constituents with the highest $p_{\mathrm{T}}$ in the event are selected. If an event has fewer constituents, NaN-padding is applied. For each constituent, the input features consist of:
  - $p_{\mathrm{T}}$: Transverse momentum.
  - $\eta$: Pseudorapidity.

– $\phi$: Azimuthal angle.

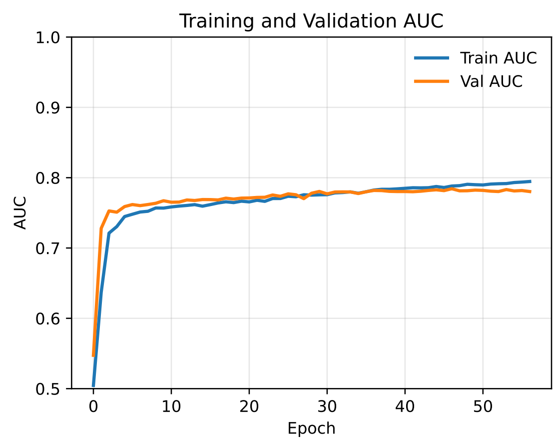
– One-hot encoding (2-dimensional vector): Identifies the constituent type as track or calorimeter tower.

This yields a feature matrix of shape $128 \times 5$ per event.

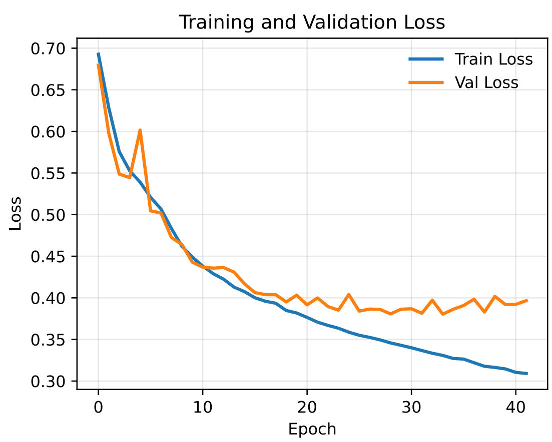
### 7.3 Training curves

The Particle Transformer models are trained using 5-fold cross-validation with an early stopping mechanism. Figure 11 presents the training and validation curves (Loss and AUC) for both the high-level and low-level feature training on the EW $W^{\pm}W^{\pm}$ vs. Bkg task.

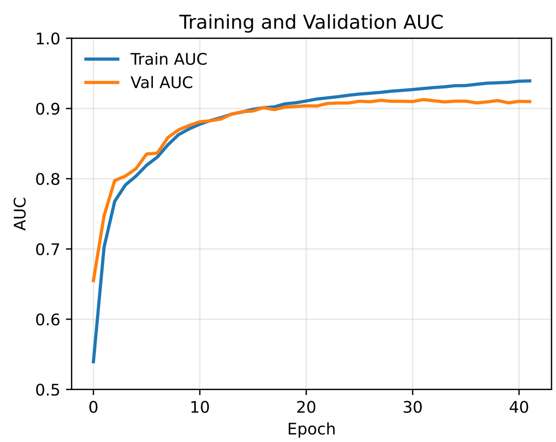

(a) Loss (High-Level features)




(b) AUC (High-Level features)




(c) Loss (Low-Level features)




(d) AUC (Low-Level features)

Figure 11: Training and validation curves for the ParT trained with high-level and low-level feature sets on the EW $W^{\pm}W^{\pm}$ vs. Bkg task.

## 7.4 Performance comparison

To quantitatively compare the classification performance of the ParT against the baseline Dense-NN, we evaluate the test AUC across three distinct training tasks: the inclusive classification task (EW $W^{\pm}W^{\pm}$ vs. Background) and the two polarization-specific tasks (LL vs. TX and LX vs. TT). Table 17 summarizes the final test AUC results.

Table 17: Comparison of the final test AUC scores between the baseline Dense-NN and the ParT utilizing high-level and low-level features across the three training tasks. The values represent the mean AUC score and its standard deviation across the 5 independent folds.

| Classification Task | Dense-NN (Baseline) | ParT (High-Level) | ParT (Low-Level) |
|---|---|---|---|
| EW $W^{\pm}W^{\pm}$ vs. Bkg | $0.806 \pm 0.001$ | $0.779 \pm 0.001$ | $0.933 \pm 0.016$ |
| LL vs. TX | $0.839 \pm 0.001$ | $0.823 \pm 0.001$ | $0.819 \pm 0.001$ |
| LX vs. TT | $0.781 \pm 0.001$ | $0.772 \pm 0.001$ | $0.763 \pm 0.003$ |

As shown in Table 17, for the polarization tasks (LL vs. TX and LX vs. TT), the performance of the ParT is comparable to or slightly lower than the baseline Dense-NN. However, for the inclusive EW $W^{\pm}W^{\pm}$ vs. Bkg classification, the low-level ParT configuration achieves a high test AUC of $0.933 \pm 0.016$, significantly outperforming both the high-level ParT ($0.779 \pm 0.001$) and the baseline Dense-NN ($0.806 \pm 0.001$).

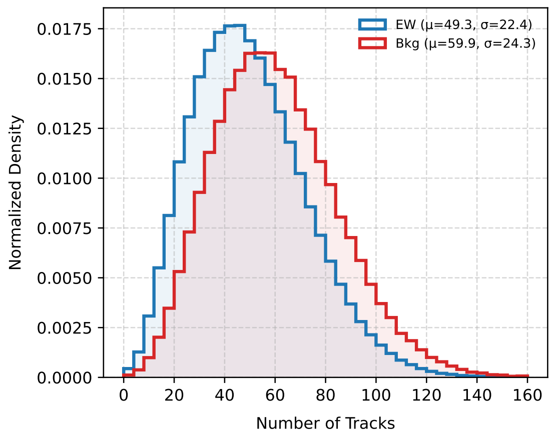
This improvement indicates that the low-level model may utilize the total count of constituent particles to distinguish the electroweak signal from the background. Figure 12 presents the distributions of the number of tracks and calorimeter towers for both EW $W^{\pm}W^{\pm}$ and background events. The figure illustrates a systematic shift toward higher constituent multiplicities in background events compared to the electroweak signal.

Figure 13 also presents the distributions for EW $W^{\pm}W^{\pm}$ with different polarizations. The figure demonstrates similar constituent multiplicities in LL, LT, and TT processes. This shows a reason why the ParT with low-level inputs didn't outperform the high-level case, since the NN can not simply use the multiplicity to distinguish them.

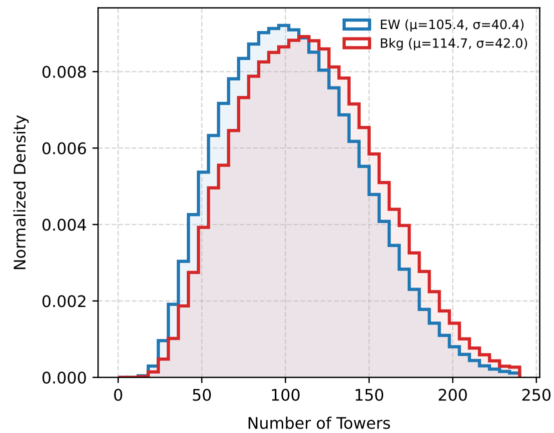
## 7.5 Training Hyperparameters

We employ a lightweight configuration of the Particle Transformer model (`ParT_Light`) to prevent overfitting on our datasets. Table 18 details the training hyperparameters and optimization settings for both the high-level and low-level feature training.

The training, validation, and testing splits follow a deterministic 5-fold cross-validation scheme matching the Dense-NN setup.

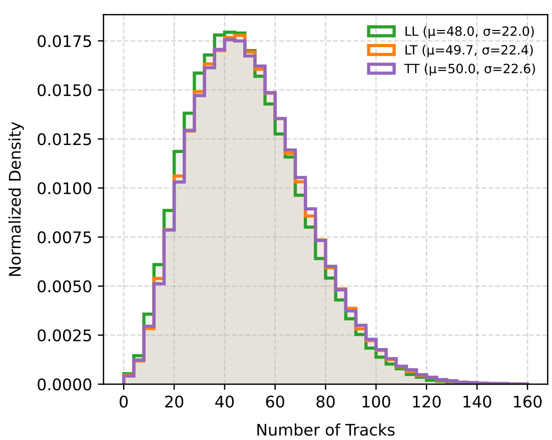



(a) Number of tracks

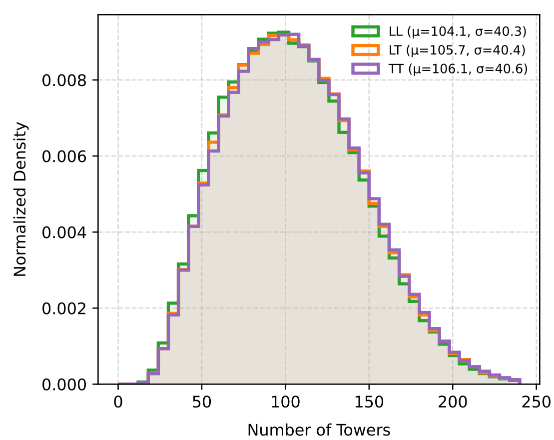



(b) Number of calorimeter towers

Figure 12: Distributions of the number of tracks and calorimeter towers for the electroweak $W^{\pm}W^{\pm}$ signal and background processes.




(a) Number of tracks




(b) Number of calorimeter towers

Figure 13: Distributions of the number of tracks and calorimeter towers for the electroweak $W^{\pm}W^{\pm}$ LL, LT, and TT processes.

Table 18: Training hyperparameters utilized in the Particle Transformer training for high-level and low-level configurations.

| Hyperparameter | High-Level Features | Low-Level Features |
|---|---|---|
| Input Sequence Length | 5 | 128 |
| Input Feature Dimension | 6 | 5 |
| Optimizer | Adam | Adam |
| Learning Rate | 0.001 | 0.001 |
| Batch Size | 2048 | 512 |
| Max Epochs | 200 | 200 |
| Early Stopping Patience | 10 epochs | 10 epochs |
| Weight Strategy | Hybrid | Hybrid |
| Random Seed | 42 | 42 |

## 7.6 Refined Low-Level Features

Since the original low-level configuration did not yield better results for the polarization separation tasks (LL vs. TX and LX vs. TT), we investigate refined low-level input feature configurations. Here, we focus on expanding the one-hot encoding representation by dividing the constituent properties into tag and type categories:

- **Type Features (4 channels):** Identifies the parent object associations of the constituent: `Track`, `Tower`, `Electron`, or `Muon`.
- **Tag Features (5 channels):** Identifies the detector reconstruction type of the constituent: `Jet1`, `Jet2`, `lepton1`, `lepton2`, or `others`.

Combining both tag and type categories yields a 9-dimensional one-hot encoding vector. Together with the 3 coordinate features ($p_{\mathrm{T}}$, $\eta$, $\phi$), the total input feature dimension per constituent is expanded to 12.

To identify which components provide the most significant classification improvement, we compare the training performance across different input settings:

1. The original low-level configuration (2 channels).
2. The combined tag and type configuration (9 channels).
3. The type-only configuration (4 channels).
4. The tag-only configuration (5 channels).

Table 19 summarizes the test AUC results for these configurations across the three training tasks.

Table 19: Comparison of the final test AUC scores for the ParT utilizing different low-level input configurations. The first row lists the baseline original low-level configuration (2 channels), while the subsequent rows present the 9-channel, 4-channel (type), and 5-channel (tag) configurations.

| Input Configuration | EW $W^{\pm}W^{\pm}$ vs. Bkg | LL vs. TX | LX vs. TT |
|---|---|---|---|
| Track & Tower | $0.933 \pm 0.016$ | $0.819 \pm 0.001$ | $0.763 \pm 0.003$ |
| Tag & Type | 0.9433 | 0.8253 | 0.7710 |
| Tag Only | 0.9314 | 0.8242 | 0.7692 |
| Type Only | 0.9458 | 0.8270 | 0.7706 |

## 7.7 Hyperparameter Tuning and Architecture Studies

To ensure a fair comparison across model configurations, we standardize the pairwise interaction features by using only kinematic angular and spatial differences: $\Delta\eta$, $\Delta\phi$, and $\Delta R = \sqrt{(\Delta\eta)^2 + (\Delta\phi)^2}$.

We test four distinct model architecture hyperparameter configurations: `Micro`, `Light`, `Deep`, and `Wide`. These configurations primarily alter the depth and width of the transformer, including the number of particle attention blocks, class attention blocks, attention heads, embedding dimensions, and feed-forward network hidden dimensions. Table 20 summarizes the detailed hyperparameters of these four experimental setups.

Table 20: Comparison of model architecture hyperparameters across the four experimental ParT configurations (`Micro`, `Light`, `Deep`, and `Wide`).

| Hyperparameter | Micro | Light | Deep | Wide |
|---|---|---|---|---|
| Particle Attention Blocks ($L$) | 2 | 3 | 6 | 3 |
| Class Attention Blocks | 1 | 1 | 2 | 1 |
| Attention Heads ($H$) | 4 | 4 | 8 | 8 |
| Embedding Dimensions | [32, 128, 32] | [64, 256, 64] | [64, 512, 64] | [64, 256, 128] |
| FC Hidden Dimension | 128 | 256 | 512 | 512 |
| Dropout | 0.1 | 0.1 | 0.1 | 0.1 |

Table 21 presents the training performance across the four model configurations for the three classification tasks with high-level input features.

Table 21: Comparison of test AUC results for the four ParT hyperparameter configurations across the classification tasks.

| Classification Task | Micro | Light | Deep | Wide |
|---|---|---|---|---|
| EW $W^{\pm}W^{\pm}$ vs. Bkg | 0.778 | 0.775 | 0.773 | 0.765 |
| LL vs. TX | 0.826 | 0.823 | 0.825 | 0.823 |
| LX vs. TT | 0.774 | 0.772 | 0.772 | 0.771 |

In addition to architectural modifications, we conduct an investigation into the impact of input feature choice by incorporating the mass ($m$) feature into the input representation. Table 22 details the classification test AUC performance across the four model configurations when the mass feature is included in the training.

Table 22: Comparison of test AUC results for the four ParT hyperparameter configurations across the classification tasks with the mass feature ($m$).

| Classification Task | Micro | Light | Deep | Wide |
|---|---|---|---|---|
| EW $W^{\pm}W^{\pm}$ vs. Bkg | 0.788 | 0.780 | 0.783 | 0.780 |
| LL vs. TX | 0.824 | 0.823 | 0.822 | 0.821 |
| LX vs. TT | 0.771 | 0.770 | 0.770 | 0.769 |

As shown in Table 21 and Table 22, the lighter `Micro` model achieves slightly superior performance compared to larger architectures, indicating that smaller models are sufficient to prevent overfitting on these high-level features. Furthermore, incorporating the mass feature yields a modest improvement in the inclusive EW $W^{\pm}W^{\pm}$ vs. Bkg classification while having negligible impact on polarization discrimination. Overall, these variations across model architectures and mass feature inclusion remain relatively small.

## 7.8 Disentangled Kinematic and Type Feature Embedding

In this section, kinematic features and constituent type one-hot encodings are projected through two independent embedding branches before being concatenated:

$$x^0 = [\text{Embed}_{\text{kin}}(X_{\text{kin}})\ ;\ \text{Embed}_{\text{type}}(X_{\text{type}})] \in \mathbb{R}^{N\times d}, \tag{21}$$

where the kinematic projection maps to $d_{\text{kin}} = \frac{3}{4}d$ and the type projection maps to $d_{\text{type}} = \frac{1}{4}d$.

Table 23 summarizes the classification test AUC performance across the four model configurations using this disentangled embedding scheme.

Table 23: Comparison of test AUC results for the four ParT hyperparameter configurations using separate kinematic and type embedding spaces across classification tasks.

| Classification Task | Micro | Light | Deep | Wide |
|---|---|---|---|---|
| EW $W^{\pm}W^{\pm}$ vs. Bkg | 0.749 | 0.753 | 0.753 | 0.751 |
| LL vs. TX | 0.797 | 0.796 | 0.798 | 0.797 |
| LX vs. TT | 0.752 | 0.754 | 0.751 | 0.752 |

As observed in Table 23, adopting this disentangled embedding scheme leads to a noticeable degradation in classification performance across all three tasks and model architectures. For instance, the test AUC for the inclusive EW $W^{\pm}W^{\pm}$ vs. Bkg task drops from $\sim 0.78$ to $\sim 0.75$, and similar declines are observed in the polarization separation tasks. This indicates that artificially isolating the kinematic and constituent type projections hinders the model's ability to learn rich joint feature representations, ultimately leading to inferior performance.

## 7.9 Impact of Input Feature Standardization

To evaluate whether feature scaling affects model convergence and discriminative capability, we conduct an investigation by applying `StandardScaler` to the continuous input features (fitting on the training fold and transforming the validation and test sets). Table 24 summarizes the test AUC scores across the four model configurations when feature standardization is applied.

Table 24: Comparison of test AUC results for the four ParT hyperparameter configurations when continuous input features are standardized using `StandardScaler`.

| Classification Task | Micro | Light | Deep | Wide |
|---|---|---|---|---|
| EW $W^{\pm}W^{\pm}$ vs. Bkg | 0.768 | 0.767 | 0.768 | 0.765 |
| LL vs. TX | 0.816 | 0.815 | 0.817 | 0.815 |
| LX vs. TT | 0.765 | 0.764 | 0.766 | 0.764 |

Comparing Table 24 with the unstandardized baseline results in Table 21, applying `StandardScaler` consistently degrades the test AUC performance by approximately 0.01 across all three classification tasks and across all model architectures. Since the internal embedding MLPs and LayerNorm layers in ParT handle feature scaling, pre-standardization introduces an unnecessary constraint that leads to slightly degraded classification performance.

## 7.10 5-Fold Cross-Validation Benchmark with Aligned Settings

To perform a fair comparison, we align the ParT training setup with the same settings.

### 7.10.1 Aligned Training Configurations

The aligned experimental settings implemented in the 5-fold cross-validation workflow are defined as follows:

- **Hybrid Weighting & Class Balancing:** Events are weighted using the Hybrid Strategy (combining physical cross-section yield with inverse event count $1/N_{\text{gen}}$). Global class balancing ($\sum w_{\text{sig}} = \sum w_{\text{bkg}}$) is enforced to prevent the loss function from favoring the majority class.

- **Model Architecture:** We employ the lightweight `ParT_Light` model structure, consisting of $L = 3$ particle attention blocks, 1 class attention block, $H = 4$ attention heads, embedding dimensions $[64, 256, 64]$, FC hidden dimension 256, and a dropout rate of 0.1.

- **Feature Preprocessing:** Transverse momentum $p_{\text{T}}$ features are transformed into logarithmic scale $\ln(p_{\text{T}})$ to enhance numerical stability.

- **Type Embedding:** We concatenate the input features with the type information first. Then, embedd them into a abstract latent space.

- **Early Stopping:** An early stopping patience of 10 epochs monitoring the validation AUC is applied to prevent overfitting.

### 7.10.2 5-Fold Performance Results

Table 25 details the 5-fold cross-validation results for the polarization task (LL vs. TX) evaluated on the high-level feature dataset under the aligned setup.

As shown in Table 25, the ParT model achieves a highly consistent test ROC-AUC score of $0.8204 \pm 0.0016$ across the 5 independent folds, exhibiting an low standard deviation. Furthermore, the negligible difference between the mean validation AUC (0.8205) and test AUC (0.8204) confirms that the aligned dataset split and training strategy effectively prevent data leakage and overfitting.

Table 25: Performance metrics across the 5 independent folds for the ParT model (`ParT_Light`) on the polarization task (LL vs. TX) using high-level features under the aligned setup.

| Fold Index | Training / Val / Test Events | Best Validation AUC | Test AUC |
|---|---|---|---|
| Fold 0 | 441,246 / 147,081 / 147,084 | 0.8184 | 0.8212 |
| Fold 1 | 441,244 / 147,084 / 147,083 | 0.8214 | 0.8207 |
| Fold 2 | 441,246 / 147,083 / 147,082 | 0.8184 | 0.8211 |
| Fold 3 | 441,248 / 147,082 / 147,081 | 0.8230 | 0.8215 |
| Fold 4 | 441,249 / 147,081 / 147,081 | 0.8214 | 0.8173 |
| **Mean ± Std** | — | $0.8205 \pm 0.0019$ | $0.8204 \pm 0.0016$ |

## 7.11 Comparative Analysis: Low-Level vs. High-Level Training

To evaluate whether detector-level constituents provide superior discrimination capability over reconstructed objects, we conduct a comparative analysis using low-level feature training.

### 7.11.1 Low-level training setup

The low-level training pipeline is configured as follows:

- **Input Constituents:** Detector-level tracks and calorimeter towers are extracted for each event.
- **Sequence Length & Padding:** To maintain a uniform sequence dimension for the Transformer, a maximum of $N_{\text{max}} = 128$ highest-$p_{\text{T}}$ constituents per event are selected. Events with fewer constituents are padded with zero vectors.
- **Feature Dimension (2-channel):** Each constituent is represented by 5 features: continuous spatial kinematics $(p_{\text{T}}, \eta, \phi)$ where $p_{\text{T}}$ is log-scaled, and a 2-dimensional one-hot encoding indicating constituent type (`Track` or `Calorimeter Tower`).
- **Hyperparameters:** The model uses the `ParT_Light` architecture with $N_{\text{max}} = 128$. The batch size is set to 512. The Adam optimizer (lr $= 10^{-3}$) and early stopping (patience $= 10$) settings match the high-level benchmark.

### 7.11.2 Performance comparison and results

Table 26 presents the test ROC-AUC performance comparison between the baseline Dense-NN, High-Level ParT (5-fold ensemble mean), and Low-Level ParT.

Table 26: Comparison of test AUC performance between the baseline Dense-NN, High-Level ParT, and Low-Level ParT across the three classification tasks.

| Classification Task | Dense-NN (Baseline) | ParT (High-Level) (5-Fold Mean) | ParT (Low-Level) Fold 0 |
|---|---|---|---|
| EW $W^{\pm}W^{\pm}$ vs. Bkg | 0.806 | 0.779 | **0.9426** |
| LL vs. TX (LL vs. LT+TT) | **0.839** | 0.8204 | 0.8047 |
| LX vs. TT (LL+LT vs. TT) | **0.781** | 0.772 | 0.7550 |

The results in Table 26 reveal two distinct behaviors:

1. **Significant Improvement in Inclusive EW $W^{\pm}W^{\pm}$ vs. Bkg:** For the inclusive signal vs. background task, the low-level ParT achieves an outstanding test AUC of **0.9426**. This substantial gain is driven by constituent multiplicity. Background processes feature higher track and tower counts due to intense QCD radiation and color reconnection, as shown in Figure 12. The ParT attention mechanism effectively leverages these low-level global multiplicity signatures.

2. **Limited Performance in Polarization Separation (LL vs. TX & LX vs. TT):** Conversely, for polarization discrimination tasks, low-level ParT is slightly lower than high-level ParT and baseline Dense-NN. This limitation stems from the physical nature of polarization states: LL, LT, and TT processes are all electroweak VBS signals with identical production channels, resulting in nearly identical constituent multiplicity distributions (Figure 13).

## 7.12 Event Scaling and Sample Size Dependence

To evaluate the impact of training sample size on the classification performance of the ParT, we conduct an event scaling experiment using the High-Level input feature set. The total Monte Carlo dataset is subsampled into five representative fractions: 100%, 50%, 25%, 10%, and 5%.

### 7.12.1 Experimental Setup and Subsampling Strategy

The scaling experiment is evaluated on Fold 0 across the three classification tasks: inclusive EW $W^{\pm}W^{\pm}$ vs. Bkg, LL vs. TX, and LX vs. TT. Models are trained using the `ParT_Light` architecture with logarithmic $p_{\mathrm{T}}$ scaling, hybrid event weighting, and an early stopping patience of 10 epochs.

To eliminate random fluctuations introduced by sampling different event populations, the subsampling procedure enforces a nested subset property. Smaller sample fractions are constructed as strict subsets of larger fractions. Consequently, performance variations across sample sizes strictly reflect the influence of available training statistics.

### 7.12.2 Performance Scaling and Physical Insights

Figure 14 presents the test AUC scores evaluated on Fold 0 as a function of the total event count across the three classification tasks.

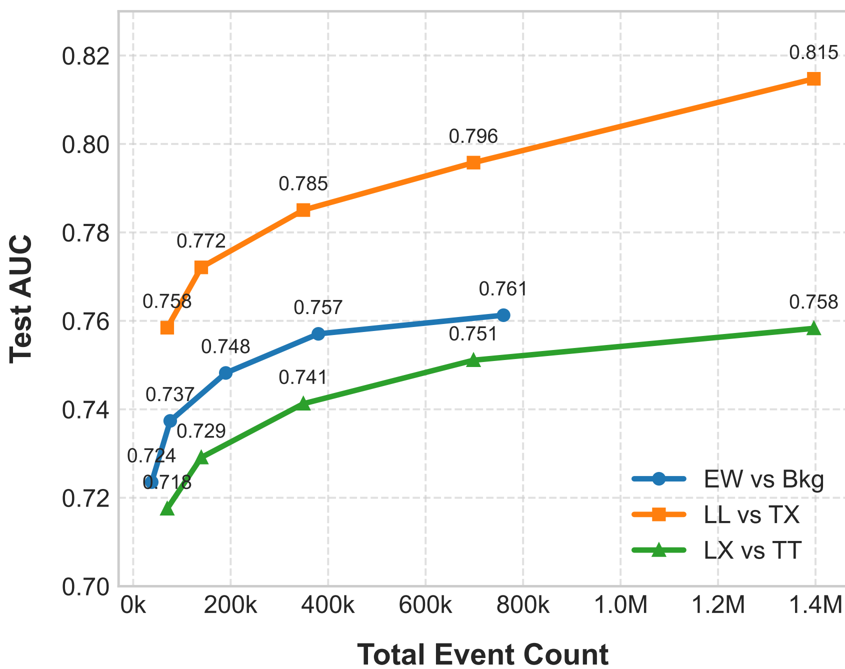



Figure 14: Test AUC performance as a function of total event count for the three classification tasks evaluated on Fold 0 using the High-Level ParT model.

The results exhibit distinct scaling behaviors across the classification tasks:

- **High Sensitivity in Polarization Tasks (LL vs. TX and LX vs. TT):** Polarization separation demonstrates a strong dependence on dataset size. For the LL vs. TX task, the test AUC increases significantly from 0.7585 at 5% sample fraction to 0.8147 at 100%, representing an absolute gain of +5.62%. Similarly, for the LX vs. TT task, the test AUC grows from 0.7175 to 0.7583 (+4.08%). Physically, the polarization states (LL, LT, TT) share identical final-state particles and production channels, differing only in subtle kinematic angular correlations. The neural network requires substantial statistics to extract these subtle high-level features effectively.

- **Steady Growth in Inclusive Classification (EW vs. Bkg):** For the inclusive signal vs. background task, the test AUC exhibits a steady improvement from 0.7235 at 5% to 0.7613 at 100%, yielding a +3.78% increase.

#### 7.12.3 Saturation Assessment

As shown in Figure 14, the AUC trajectory for the polarization tasks maintains a upward slope up to the 1.40M events, indicating that performance has not yet reached saturation. This demonstrates that expanding the simulated Monte Carlo sample size offers substantial potential for improving polarization measurement sensitivity.

# References


[1] G. Aad *et al.*, "Evidence for Longitudinally Polarized W Bosons in the Electroweak Production of Same-Sign W Boson Pairs in Association with Two Jets in pp Collisions at s=13 TeV with the ATLAS Detector," *Phys. Rev. Lett.*, vol. 135, no. 11, p. 111802, 2025.

[2] J. Alwall, R. Frederix, S. Frixione, V. Hirschi, F. Maltoni, O. Mattelaer, H. S. Shao, T. Stelzer, P. Torrielli, and M. Zaro, "The automated computation of tree-level and next-to-leading order differential cross sections, and their matching to parton shower simulations," *JHEP*, vol. 07, p. 079, 2014.

[3] T. Sjöstrand, S. Ask, J. R. Christiansen, R. Corke, N. Desai, P. Ilten, S. Mrenna, S. Prestel, C. O. Rasmussen, and P. Z. Skands, "An introduction to PYTHIA 8.2," *Comput. Phys. Commun.*, vol. 191, pp. 159–177, 2015.

[4] J. de Favereau, C. Delaere, P. Demin, A. Giammanco, V. Lemaître, A. Mertens, and M. Selvaggi, "DELPHES 3, A modular framework for fast simulation of a generic collider experiment," *JHEP*, vol. 02, p. 057, 2014.

[5] M. Cacciari, G. P. Salam, and G. Soyez, "The anti-$k_t$ jet clustering algorithm," *JHEP*, vol. 04, p. 063, 2008.

[6] M. Cacciari, G. P. Salam, and G. Soyez, "FastJet User Manual," *Eur. Phys. J. C*, vol. 72, p. 1896, 2012.

[7] M. Stange, M. Kobel, F. Siegert, and K. Potamianos, *Longitudinal W± Boson Polarization in Same-charged W±W± Scattering in the ATLAS Experiment*. Verlagsname nicht ermittelbar, 2025.

[8] D. Buarque Franzosi, O. Mattelaer, R. Ruiz, and S. Shil, "Automated predictions from polarized matrix elements," *JHEP*, vol. 04, p. 082, 2020.

[9] E. Bothmann *et al.*, "Event generation with Sherpa 3," *JHEP*, vol. 12, p. 156, 2024.

[10] R. D. Ball *et al.*, "Parton distributions with LHC data," *Nucl. Phys. B*, vol. 867, pp. 244–289, 2013.

[11] R. D. Ball *et al.*, "The PDF4LHC21 combination of global PDF fits for the LHC Run III," *J. Phys. G*, vol. 49, no. 8, p. 080501, 2022.

[12] G. Aad *et al.*, "Measurement and interpretation of same-sign W boson pair production in association with two jets in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector," *JHEP*, vol. 04, p. 026, 2024.

[13] D. L. Rainwater, R. Szalapski, and D. Zeppenfeld, "Probing color singlet exchange in $Z$ + two jet events at the CERN LHC," *Phys. Rev. D*, vol. 54, pp. 6680–6689, 1996.

[14] P. Ramachandran, B. Zoph, and Q. V. Le, "Searching for activation functions," *CoRR*, vol. abs/1710.05941, 2017.

[15] J. Ansel, E. Yang, H. He, N. Gimelshein, A. Jain, M. Voznesensky, B. Bao, P. Bell, D. Berard, E. Burovski, G. Chauhan, A. Chourdia, W. Constable, A. Desmaison, Z. DeVito, E. Ellison, W. Feng, J. Gong, M. Gschwind, B. Hirsh, S. Huang, K. Kalambarkar, L. Kirsch, M. Lazos, M. Lezcano, Y. Liang, J. Liang, Y. Lu, C. K. Luk, B. Maher, Y. Pan, C. Puhrsch, M. Reso, M. Saroufim, M. Y. Siraichi, H. Suk, S. Zhang, M. Suo, P. Tillet, X. Zhao, E. Wang, K. Zhou, R. Zou, X. Wang, A. Mathews, W. Wen, G. Chanan, P. Wu, and S. Chintala, "Pytorch 2: Faster machine learning through dynamic python bytecode transformation and graph compilation," in *Proceedings of the*

*29th ACM International Conference on Architectural Support for Programming Languages and Operating Systems, Volume 2*, ASPLOS '24, (New York, NY, USA), p. 929–947, Association for Computing Machinery, 2024.

[16] H. Qu and L. Gouskos, "ParticleNet: Jet Tagging via Particle Clouds," *Phys. Rev. D*, vol. 101, no. 5, p. 056019, 2020.

[17] H. Qu, C. Li, and S. Qian, "Particle Transformer for Jet Tagging," 2 2022.

[18] E. M. Metodiev, B. Nachman, and J. Thaler, "Classification without labels: Learning from mixed samples in high energy physics," *JHEP*, vol. 10, p. 174, 2017.

[19] E. M. Metodiev and J. Thaler, "Jet Topics: Disentangling Quarks and Gluons at Colliders," *Phys. Rev. Lett.*, vol. 120, no. 24, p. 241602, 2018.

[20] B. M. Dillon, G. Kasieczka, H. Olischlager, T. Plehn, P. Sorrenson, and L. Vogel, "Symmetries, safety, and self-supervision," *SciPost Phys.*, vol. 12, no. 6, p. 188, 2022.

[21] X. Wang, "Measurements of vector-boson scattering with the ATLAS experiment," 2024.

[22] S. Folgueras, "Multiboson and vbs measurements in atlas and cms," 2026.

[23] M. Hu, *Precision Measurements of Vector Boson Scattering Production and Searches for Charged Higgs Bosons at the Large Hadron Collider.* PhD thesis, Massachusetts Institute of Technology, 2022.

[24] G. Aad *et al.*, "Search for diboson resonances in hadronic final states in 139 $\text{fb}^{-1}$ of $pp$ collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector," *JHEP*, vol. 09, p. 091, 2019. [Erratum: JHEP 06, 042 (2020)].

[25] A. Tumasyan *et al.*, "Search for new heavy resonances decaying to WW, WZ, ZZ, WH, or ZH boson pairs in the all-jets final state in proton-proton collisions at $\sqrt{s} = 13$ TeV," *Phys. Lett. B*, vol. 844, p. 137813, 2023.

[26] R. L. Delgado, A. Dobado, F. J. Llanes-Estrada, and J. J. Sanz-Cillero, "Dynamical vector resonances from the echl in vbs at the lhc: the ww case," *The European Physical Journal C*, vol. 79, Nov. 2019.

[27] H. Li, M. Stamenkovic, A. Shmakov, M. Fenton, D. S.-C. Chao, K. M. White, C. Mikkelsen, J. Mitić, C. Mantilla Suarez, M. Quinnan, G. Landsberg, H. Newman, P. Baldi, D. Whiteson, and J. Duarte, "Reconstruction of boosted and resolved

multi-Higgs-boson events with symmetry-preserving attention networks," *JHEP*, vol. 11, p. 119, 2025.